

Technical change and health: Evidence from the reduction of sugarcane burning in Brazil*

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Abstract

São Paulo, the major sugarcane-producing state in Brazil, commands over a fifth of the world's sugarcane production. Between the late 2000s and the mid-2010s, the state replaced manual pre-harvest burning with mechanical harvesting across nearly its entire sugarcane area – a shift we use to study its consequences for public health in São Paulo's municipalities. Using a difference-in-differences design that exploits variation in municipal sugarcane land share, we find that greater exposure to the technical change reduced respiratory hospitalizations. Circulatory hospitalizations also decline, but this decline cannot be separated from a convergence already under way before the transition. We further document reductions in pollution-linked congenital anomalies. Mortality from respiratory and circulatory causes shows no detectable effect, among adults or among children under five. The pattern of the effect points to the smoke channel: fire incidence and particulate matter fall only during the harvest season, respiratory hospitalization reductions are larger during those same months, and municipalities downwind of sugarcane areas benefit more.

Keywords: Sugarcane Burning; Technical Change; Health Outcomes; Air Pollution.

JEL Codes: Q52, Q55, I18.

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1 Introduction

Agricultural fires have several drawbacks during and after burning, including alterations to soil characteristics and significant local air pollution ([Andreae and Merlet, 2001](#); [Santín and Doerr, 2016](#); [Guo, 2021](#)). These fires translate into population health costs: higher infant mortality ([Pullabhotla et al., 2023](#)), elevated hypertension risk ([Pullabhotla and Souza, 2022](#)), and greater incidence of acute respiratory infection ([Chakrabarti et al., 2019](#)). Despite this evidence, fires remain widely used in agriculture because they reduce costs associated with clearing bushes, crop residue management, and controlling invasive plant species ([Levine, 1991](#); [Radosevich et al., 2007](#)). This cost reduction is particularly important for smallholders, but fires are also common in large-scale agriculture – *e.g.*, pre-harvest sugarcane burning in Brazil ([Uriarte et al., 2009](#)) and wheat post-harvest in China, India, Pakistan, Nepal, and Bangladesh ([Lin and Begho, 2022](#); [Li et al., 2008](#)).

São Paulo’s elimination of pre-harvest sugarcane burning is, to our knowledge, one of the largest-scale cases of agricultural fire elimination to date. It is also one of the best settings to study the relationship between agricultural fire and health. The setting is concentrated and intensive. Brazil dominates the world sugarcane market, accounting for 39% of global production in 2024 ([FAO, 2025](#)). São Paulo state sits at its center: as Brazil’s primary producer, it contributes 58% of the nation’s output – over a fifth (21%) of world production – while allocating around 20% of its territory to sugarcane by 2010 ([IBGE, 2024](#)). The technological shift was also fast and complete. Driven by a mix of state environmental regulation, a voluntary industry agreement, rising rural wages ([Davis, 2017](#)), and export-market pressure for sustainable production, the share of sugarcane harvested mechanically rose from roughly one third in 2007 to over 95% by 2015 ([Conab, 2024](#)). Finally, because the transition occurred within a single, comparably governed jurisdiction, we can hold constant state-level institutions, regulations, and other common factors while comparing municipalities with sharply different exposure to the technical change.

We exploit the uneven distribution of sugarcane production across São Paulo’s 645 municipalities to quantify exposure to the technical change. While half of all municipalities allocate minimal or no land to sugarcane cultivation, the other half dedicates at least 5% of their territory to this crop, and the top decile allocates over 45% ([IBGE, 2024](#)). Individuals in municipalities with more land allocated to sugarcane were more exposed to sugarcane smoke before the transition and stand to benefit more from its elimination. This setting is well suited for a difference-in-differences design with continuous treatment, where the state-level timing of the burning reduction interacts with the municipality-level sugarcane land

share to generate variation in exposure to the technical change that is plausibly exogenous to local health trends ([Callaway et al., forthcoming](#)).

We combine several data sources. Our primary health outcomes come from three Ministry of Health registries: hospital admissions for respiratory and cardiovascular conditions, assigned to the patient’s municipality of residence, from the Hospital Information System (SIH/SUS) over 2008–2021; birth records – including weight, gestational age, detailed anomaly information, and maternal characteristics – from the Live Birth Information System (Sinasc) over 2002–2023; and deaths attributed to respiratory and cardiovascular causes from the Mortality Information System (SIM) over 2002–2023. Our treatment variable is the pre-2011 average share of municipal land area dedicated to sugarcane, constructed from IBGE’s Municipal Agricultural Survey (PAM). Results are materially unchanged when this average is instead computed over other pre-treatment windows. To document the pollution channel, we use daily PM_{10} and $PM_{2.5}$ estimates from the Copernicus Atmosphere Monitoring Service (CAMS) global reanalysis, aggregated to the municipality level ([Saldanha et al., 2026b,a](#)), and verify fire incidence by cross-referencing satellite burned-area data (Modis) with sugarcane land cover (MapBiomas). To test whether health improvements are larger in municipalities exposed to smoke carried by prevailing winds, we construct a downwind exposure measure from Nasa Power (Merra-2) wind direction for the harvest months.

The transition to mechanical harvesting reduced respiratory hospitalizations: a one-standard-deviation increase in exposure ($SD = 0.19$) reduces them by 7.3%. Circulatory hospitalizations fall by 3.4%, but circulatory outcomes in high-sugarcane municipalities were already converging before 2011. The circulatory estimates cannot be distinguished from a continuation of that pre-existing convergence. Birthweight outcomes show no meaningful effect: very low birthweight and continuous birthweight (levels and logs) are all statistically indistinguishable from zero. A small positive coefficient at the low-birthweight threshold is not corroborated by either the mean or a more severe threshold. We do, however, document reductions in congenital anomalies – and crucially, the effect is concentrated among anomaly types with established biological pathways to particulate matter exposure. A one-standard-deviation increase in exposure reduces the incidence of the pollution-linked anomaly index – comprising neural tube defects, orofacial clefts, and cardiac malformations – by 25.7% relative to baseline, driven primarily by cardiac malformations (–39.5%) and neural tube defects (–12.3%). Mortality from respiratory and circulatory causes shows no detectable change, either among adults or among children under five.

One genuine question is whether these health improvements occurred through the smoke channel, and we confirm this using several pieces of evidence. Mechanical harvesting reduced PM_{10} and $PM_{2.5}$ concentrations during the harvest season (July–September), with no clear effect in the off-season. Fire incidence on sugarcane fields also fell sharply during the harvest season, providing direct evidence that the reduction in particulate matter coincides with less burning rather than long-run improvements in air quality. Municipalities downwind of sugarcane areas experienced larger health improvements. Hospitalizations for injuries and falls, which have no plausible link to air pollution, show no response. A continuous treatment measure replaces the binary post-2010 indicator with the observed fraction of sugarcane-burning emissions already eliminated in each year, yielding effects that are very close to the baseline estimates.

This evidence matters beyond São Paulo because several Brazilian states continue to rely on pre-harvest burning. In the Northeast, where only one-third of the sugarcane harvest is mechanized (Conab, 2024), the practice remains widespread. The use of fire in sugarcane cultivation remains common around the world as well – including in Florida (Nowell et al., 2022)¹, India (Ravish and Chaudhry, 2025), and Thailand (Phairuang et al., 2019; Prasertsri et al., 2025). Our findings thus speak to a broader tension in agricultural production: practices that reduce short-run costs but impose lasting health burdens on surrounding populations. We estimate a lower bound for these costs at R\$ 118 million (USD 30 million), only considering avoided respiratory hospitalizations and congenital anomalies over a decade. It excludes several costs that we cannot measure, such as outpatient visits, productivity losses, the lifetime social protection costs associated with severe birth defects, other pollution-related diagnoses beyond our core outcomes, and cases in which pollution lengthens or raises the cost of hospitalizations for unrelated causes. This is necessarily a lower-bound estimate of the total social cost. Still, quantifying at least part of these costs matters because leaving them entirely unmeasured would make the transition toward fire-free harvesting policies harder to justify.

The closest work to ours quantifies the health costs of agricultural fires specifically (Guo, 2021; Pullabhotla and Souza, 2022; Pullabhotla et al., 2023; Chakrabarti et al., 2019); we ask what happens when such fires are eliminated at scale, rather than merely reduced. More broadly, our estimates speak to the literature on the health effects of air pollution (Burnett et al., 2018; Turner et al., 2020) and to studies examining the health impacts of

¹Florida presents a striking case: pre-harvest burning persists despite the harvest being fully mechanized. Growers argue that burning reduces costs and accelerates the harvest by eliminating leaf trash before mechanical cutting. Critics counter that the practice continues primarily because of the political influence of the sugar industry at the state level (Nowell et al., 2022).

environmental regulations and agreements (Isen et al., 2017; McCarty et al., 2007). Prior research has linked air pollution exposure to adverse birth outcomes (Currie and Schmieder, 2009; Sram et al., 2013; Greenstone and Hanna, 2014; Van Rossem et al., 2015), respiratory illnesses (Mo et al., 2018; Simkovich et al., 2019; Tran et al., 2023), and circulatory diseases (Franchini and Mannucci, 2007; Lee et al., 2014; Franklin et al., 2015). Agricultural burning is a pollution source that has received comparatively little attention given its scale, and our estimates help fill that gap. Previous research on pre-harvest sugarcane burning in Brazil has focused on emissions, air pollution, and air quality, without linking these effects to health outcomes (Valente and Laurini, 2021; Cristale et al., 2012; Souto-Oliveira et al., 2023; França et al., 2014). Some studies examine the relationship between sugarcane smoke and health in specific municipalities (Arbex et al., 2000; Cançado et al., 2006) or among workers directly exposed (Prado et al., 2012; Abreu et al., 2011; Ribeiro and Ficarelli, 2010). Differently, we estimate effects for the general population, rather than for a single locality or an occupationally exposed subgroup.

The closest antecedents in the Brazilian sugarcane context are Rangel and Vogl (2019) and Chagas et al. (2016). Rangel and Vogl (2019) examines the impact of agricultural fires on birth outcomes, exploiting daily variation in wind direction and fire location to isolate smoke exposure. Upwind fires led to lower birthweight and shorter gestational age, while non-upwind fires had only mild effects. Chagas et al. (2016) estimates the effects of sugarcane production on respiratory hospitalizations, introducing a spatial difference-in-differences design that accounts for spillover effects onto neighboring non-producing municipalities. Our contribution differs along three main dimensions. First, we broaden the set of outcomes to include circulatory hospitalizations, infant mortality, and pollution-linked congenital anomalies. Cardiac malformations and neural tube defects have established biological pathways to particulate matter exposure (Tanwar et al., 2017; Chen et al., 2024), yet have not been studied in the sugarcane burning literature. Second, our sample period captures a sharp decline in pre-harvest burning while cultivated area remained broadly stable, allowing us to separate the reduction in burning from changes in sugarcane production that characterized earlier periods. Third, our continuous treatment measure captures differences in exposure intensity that a binary treatment based on a median sugarcane threshold cannot: municipalities allocating 6.8% and 50% of their land to sugarcane receive the same treatment status under a median split, while a municipality at 6.6% falls into the control group. Our design instead weights municipalities by their pre-treatment sugarcane land share, so that identification varies with actual exposure intensity.

2 Background

2.1 Sugarcane production in Brazil

Brazil is the world's leading producer of sugarcane, harvesting approximately 760 million tonnes in 2024 – 40% of global output (Figure A-1) (FAO, 2025). Domestically, sugarcane is split roughly evenly between two uses: sugar and ethanol (Conab, 2024). This balance shifts at the margin in response to market conditions – *e.g.*, the sugar share rises when international prices are favorable to producers, and the ethanol share expands when policies that mandate or subsidize biofuel consumption are tightened, or when higher international oil prices make ethanol more competitive at the pump (Goldemberg et al., 2018).

Between 2000 and 2010, Brazil's sugarcane production grew by roughly 50% – from 326 to 717 million tonnes per harvest – driven by biofuel policies that included rising ethanol mandates and preferential taxation of ethanol relative to gasoline (Cardoso et al., 2019). This expansion was channeled through two distinct ethanol markets. Anhydrous ethanol is blended directly with gasoline under a federally mandated ratio, set at 30% as of August 2025 (CNPE, 2025), and affects fuel consumption symmetrically across all regions. Hydrated ethanol, by contrast, is sold as a standalone fuel and can power any flex-fuel vehicle – which account for more than 80% of Brazil's light vehicle fleet (Anfavea, 2025). Unlike anhydrous ethanol, hydrated ethanol competes directly with gasoline at the pump, and that competition is geographically uneven: transport costs erode hydrated ethanol price advantage with distance from the sugarcane belt. In regions far from the producing areas – such as the North – consumers face prices at which hydrated ethanol is no longer competitive with gasoline, effectively suppressing demand.² Figure A-10 maps this geographic pattern.³

[Figure 1]

Three states account for 75% of the sugarcane production in Brazil: São Paulo (52%), Goiás (12%), and Minas Gerais (11%)⁴. São Paulo alone produced 350 million tonnes in 2025 – more than half of Brazil's national output of 673 million tonnes that year (IBGE, 2024)⁵ – across 4.4 million hectares. Were São Paulo a country, it would be the world's

²For a comprehensive overview of ethanol adoption in Brazil, see Fontana et al. (2026).

³A further structural shift is underway: corn-based ethanol has grown rapidly and by 2026 accounted for more than 20% of total ethanol production in Brazil (Conab, 2024).

⁴Based on Conab (2024) data for the 2025/2026 harvest.

⁵FAO (2025) reports a slightly higher national total of 717 million tonnes for the same year, reflecting methodological differences in coverage. We use IBGE/PAM figures throughout for within-Brazil comparisons.

second-largest sugarcane producer, trailing only India (473 million tonnes in 2024) (FAO, 2025) and still ahead of the rest of Brazil combined (323 million tonnes). Production is also concentrated within the state: most of it comes from municipalities in the northwest, while the south contributes negligibly. Figure A-2 shows the share of municipal land allocated to sugarcane cultivation across São Paulo from 2000 to 2010, the pre-treatment period.

[Figure 2]

There is considerable heterogeneity in the intensity of sugarcane cultivation across municipalities. Among the 645 municipalities in São Paulo state, 230 have virtually no sugarcane production (less than 2% of their territory allocated to sugarcane), while 57 municipalities dedicate at least half of their land to sugarcane cultivation (see Figures A-5 and A-6 for the distribution). We exploit this spatial variation in the share of land under sugarcane cultivation as a measure of each municipality's exposure to sugarcane smoke.

2.2 *The Mechanization of Sugarcane Harvesting in São Paulo*

Figures A-7 and A-9 document the shift in São Paulo's sugarcane sector over the past two decades: harvested area expanded sharply through the late 2000s before stabilizing, while the share of mechanically harvested cane rose from roughly one-third in 2007 to near-universality by the mid-2010s.

The state government has emphasized its own role in this transition, pointing to two complementary regulatory interventions. First, the State Law No. 11.241/2002 established a gradual ban on sugarcane straw burning, with targets of 50% of harvested area by 2011, 80% by 2016, and 100% by 2021. Second, the Green Ethanol Protocol,⁶ a voluntary agreement signed in 2007 between the state government and the sugarcane industry, accelerated these deadlines to 2014 and 2017 for mechanizable and other regions, respectively, offering refineries access to preferential financing and environmentally demanding markets in exchange for early compliance.

Davis (2017) challenges the regulatory narrative, arguing that the law explains little of the mechanization observed in São Paulo. His test is geographic: if regulation were the binding constraint, mechanization should be concentrated in São Paulo – where the law applied – yet the data do not clearly support this pattern⁷. Drawing instead on sur-

⁶In Portuguese, *Protocolo Agroambiental Etanol Verde* or *Protocolo Etanol Mais Verde* after 2017.

⁷This test may be complicated by the close economic integration between the north of São Paulo and the south of Minas Gerais – the second-largest producer – which share many of the same companies, agricultural practices, and input suppliers, potentially blurring the regulatory contrast across the state border.

vey evidence, he documents that growers switched to mechanical harvesting primarily in response to rising real wages among low-skilled workers, a pressure that intensified throughout Brazil during the 2000s. This interpretation is further supported by the limited mechanization observed in poorer sugarcane-producing regions of Brazil, where the same wage pressure was absent.

A third driver operates through export market access. Sugar is heavily traded internationally, and buyers – particularly in Europe – increasingly conditioned market access on environmentally responsible production practices. Pre-harvest burning threatened the sustainability credentials of Brazilian sugar at a time when export markets were growing more sensitive to production methods. This demand-side pressure reinforced the incentives created by regulation and labor costs, particularly for mills with significant export exposure ([De Moraes and Zilberman, 2014](#)).

The relative weight of these three forces – environmental regulation, rising wages, and export market pressure – remains an open question, and settling it is beyond the scope of this paper. What is clear is that all three contributed to a transition that, by the early 2010s, had fundamentally altered how sugarcane is harvested in São Paulo.

Figure [A-3](#) documents the technical change indirectly. The total area permitted for pre-harvest burning peaked at nearly two million hectares around 2010 and fell to near zero by 2020. Yet a reduction in the area legally permitted for burning need not translate into a reduction in actual fire use if growers do not comply with the policy. Section [5](#) addresses this concern, showing that municipalities more exposed to the technical change experienced significant reductions in PM_{10} and $PM_{2.5}$ – reductions concentrated exclusively during the harvest season (July–September) and statistically indistinguishable from zero in the off-season. This seasonal pattern is difficult to reconcile with any explanation other than reduced agricultural burning.

3 Data and Empirical Strategy

3.1 Data

Our analysis combines six administrative and environmental datasets for São Paulo state.⁸ Summary statistics for all main variables are reported in Table A-6 in the Appendix.

The treatment variable is constructed from the Municipal Agricultural Production survey (PAM) maintained by IBGE (IBGE, 2024). We compute the share of each municipality's total area dedicated to sugarcane cultivation, averaged over the pre-treatment period 2000–2010. This pre-period average serves as our primary measure of each municipality's exposure intensity to the technical change. Among the 645 municipalities in São Paulo,⁹ roughly half allocate less than 5% of their territory to sugarcane, while the top decile dedicates more than 45%. Figures A-5 and A-6 illustrate how this variable is distributed.

For health outcomes, we draw on three Ministry of Health administrative registries. Hospital admissions come from the Sistema de Informação Hospitalar (SIH/SUS) via Datasus, spanning 2008–2022. We focus on admissions with principal diagnoses in ICD-10 chapters J (respiratory diseases, J00–J99) and I (circulatory diseases, I00–I99), aggregated to municipality-year cells and expressed as rates per capita. Birth records come from the Sistema de Nascidos Vivos (SINASC) via Datasus, covering approximately 12.0 million individual births in São Paulo over 2002–2023. Each record contains birthweight, a congenital anomaly indicator, birth month, maternal education, and municipality of residence. We recode the anomaly indicator from its raw form (1 = yes, 2 = no, 9 = unknown) to a binary variable, treating unknown codes as missing. We also construct a pollution-linked anomaly index (Pol3) equal to one if the birth presents at least one of: neural tube defects (Q00–Q07), orofacial clefts (Q35–Q37), or congenital cardiac malformations (Q20–Q26). Mortality records come from the Sistema de Informações sobre Mortalidade (SIM/Datasus), covering 2002–2023. We use respiratory and circulatory cause-specific deaths relative to population, for the full population and for residents aged 60 and over, and respiratory deaths relative to live births for children under five.

⁸This subsection describes the data used in the paper and, just as importantly, how to access it. Three R packages are central to this task – `ipeadatar`, `sidrar`, and `microdatasus` – which provide access to GDP and population data, PAM, and Datasus records, respectively. Google Earth Engine was used to overlay MapBiomas land cover data (to locate sugarcane fields) with Modis burned-area data.

⁹Our estimation sample contains 642 of these municipalities. PAM reports no temporary-crop area for Diadema, São Caetano do Sul, and Ilha Comprida in any year of our sample – the first two are dense, fully urbanized municipalities in the São Paulo metropolitan area and the third is a coastal island largely under environmental protection – so none has sugarcane cultivation and all three would enter the control group.

To document the smoke channel, we combine three independent data sources. Air quality data come from daily surface-level PM₁₀ and PM_{2.5} concentration estimates from the Copernicus Atmosphere Monitoring Service (CAMS) global reanalysis (EAC4 product), processed to municipality-level daily means via zonal statistics (Saldanha et al., 2026b,a), available over 2003–2024, which we aggregate to municipality-season-year cells. Fire incidence is measured by cross-referencing the MODIS MCD64A1 monthly burned-area product (Collection 6.1) with the MapBiomas Land Use and Land Cover sugarcane layer (Collection 9), both processed via Google Earth Engine at MODIS’s native 500-meter resolution – the coarser of the two inputs. This cross-referencing allows us to isolate burned area within sugarcane pixels and to compare the harvest season (July–September) against the off-season (January–March) as a placebo. Finally, state-level annual sugarcane burning emissions come from the Sistema de Estimativas de Emissões e Remoções de Gases de Efeito Estufa (SEEG) (SEEG, 2024), covering 2000–2018; Section 3.2 uses this series to construct $fase_t$, our continuous measure of treatment intensity.

The wind direction triple-difference relies on a municipality-level upwind sugarcane exposure measure constructed from Nasa Power/Merra-2 reanalysis wind data for July–September, 2002–2023. For municipality i in year t :

$$WindExp_{it} = \sum_j \frac{\max(0, \cos(\hat{w}_{it} - \angle_{i \rightarrow j}))}{dist_{ij}} \cdot Land_j, \quad dist_{ij} \leq 100 \text{ km}$$

where $\hat{w}_{it}$ is the prevailing wind direction at i , $\angle_{i \rightarrow j}$ is the bearing from i toward source municipality j , and $Land_j$ is j ’s sugarcane land share. Because wind direction varies from year to year, $WindExp_{it}$ has within-municipality time variation and is not collinear with the municipality fixed effects. We also collect population estimates from IBGE and GDP per capita from IBGE municipal accounts, used as regression weights and controls, respectively.

3.2 Empirical Strategy

Our primary design exploits the timing of the burning reduction and the spatial variation in sugarcane land use:

$$\ln(Y_{it}) = \alpha \cdot TC_{it} + \theta_i + \gamma_t + \epsilon_{it} \quad (1)$$

where Y_{it} is the health outcome of municipality i in year t ; θ_i and γ_t are municipality and year fixed effects, respectively. $TC_{it} = Land_i \times D_{t > 2010}$ is the exposure to the technical

change – the product of the pre-treatment sugarcane land share and a dummy equal to one after 2010. α is our parameter of interest. Regressions are weighted by population (or by births for birth outcomes), and standard errors are clustered at the municipality level. For individual-level birth outcomes, we estimate the analogous linear probability model (LPM), replacing municipality-year observations with individual births and adding fixed effects for maternal education, sex, and marital status. Our continuous-treatment specification follows Callaway and Sant’Anna (forthcoming) for settings with common treatment timing and continuous exposure intensity.

We augment the baseline specification with three strategies that sharpen identification. The first is a harvest season triple-difference, which exploits the fact that sugarcane burning – and thus PM exposure – is concentrated in July-September:

$$\ln Y_{itm} = \alpha \cdot TC_{it} + \beta \cdot TC_{it} \times Harvest_m + \theta_{i \times m} + \gamma_t + \varepsilon_{itm} \quad (2)$$

where $Harvest_m = 1(m \in \{7, 8, 9\})$ and $\theta_{i \times m}$ are municipality $\times$ month fixed effects that absorb local seasonal patterns. The coefficient β captures the additional effect of exposure during burning months.

The second strategy adds the time-varying $WindExp_{it}$ variable to equation (1):

$$\ln(Y_{it}) = \alpha \cdot TC_{it} + \delta \cdot WindExp_{it} + \theta_i + \gamma_t + \epsilon_{it} \quad (3)$$

testing whether municipalities downwind of sugarcane areas benefit more from the transition.

The third replaces the binary $D_{t>2010}$ indicator with $fase_t \in [0, 1]$ – the fraction of pre-2011 baseline sugarcane burning emissions already eliminated in year t , constructed from SEEG data (SEEG, 2024) – interacted with the municipal sugarcane land share:

$$\ln(Y_{it}) = \alpha \cdot (Land_i \times fase_t) + \theta_i + \gamma_t + \epsilon_{it} \quad (4)$$

Because $fase_t$ is common to all municipalities in a given year, it provides an exogenous state-level shifter that scales with the actual pace of the emissions reduction.

4 Results

4.1 Hospitalizations

Table 1 presents results for respiratory and circulatory hospitalizations. Technical change exposure reduces hospitalizations across all specifications and both disease categories, with respiratory outcomes responding more strongly than circulatory ones – consistent with the primarily airborne nature of the smoke channel. The direction of the effect is intuitive: municipalities more intensive in sugarcane production were more exposed to pre-harvest burning and, consequently, stand to benefit more from its elimination.

Specifically, for respiratory diseases (Panel A), a one-standard-deviation increase in exposure ($SD = 0.19$) is associated with a 7.3% reduction in per capita hospitalizations (0.386×0.19). The estimates are robust to controlling for per capita GDP and to excluding the COVID-19 period. Pre-COVID coefficients are slightly smaller in magnitude (-0.300 vs. -0.386), a difference of roughly 22% that remains qualitatively consistent with the full-sample results. Controlling for GDP leaves the point estimates virtually unchanged – suggesting that the health improvements are not driven by differential economic trends across municipalities.

For circulatory diseases (Panel B), a one-standard-deviation increase in exposure reduces hospitalizations by 3.4%. Across all specifications, the circulatory point estimates are considerably smaller than their respiratory counterparts – less than half their size. The circulatory estimates are fully stable across both the full sample and the pre-COVID period, and controlling for GDP leaves the point estimates essentially unchanged, as with respiratory diseases.

4.2 Birth Outcomes

Table 2 presents results from the individual-level analysis, estimated on approximately 12.6 million SINASC birth records from São Paulo (2002–2023). The specification extends the baseline design with additional fixed effects for maternal education, sex, and marital status. Birthweight outcomes show no meaningful effect. Very low birthweight (VLBW) is statistically indistinguishable from zero ($p = 0.655$), consistent with the interpretation that exposure to the technical change lies below the threshold for the most severe fetal growth restriction, and birthweight itself shows no detectable relationship with exposure, whether measured in levels (grams) or in logs ($p = 0.20$ and $p = 0.28$, respectively). Low

birthweight (LBW) – a single threshold at 2,500g – is positive and significant at the 5% level ($p = 0.048$), but we read this as noise rather than a genuine effect. The point estimate is small: at one standard deviation of exposure, it implies less than a one percent relative increase in the LBW rate (from a baseline of 9.2 per 100 births), an order of magnitude smaller than the effects we document below for congenital anomalies. More importantly, it does not corroborate: the underlying continuous outcome shows nothing, and the more severe VLBW threshold shows nothing either. A single significant coefficient at one specific cutoff, absent from both the mean and a more extreme threshold of the same variable, is the signature of a false positive rather than a systematic pattern.¹⁰

The technical change does, however, matter for developmental processes during critical windows of organogenesis. Greater exposure significantly reduces both overall congenital anomalies and the pollution-linked anomaly index. Because recorded anomaly rates in SINASC rise roughly tenfold between 2002 and 2023 – reflecting the expansion of prenatal ultrasound and fetal echocardiography access rather than a change in true incidence – we benchmark relative effects against the post-2010 mean, which is representative of the diagnostic environment during the treatment period; year fixed effects absorb this secular trend from the coefficients themselves. For the overall anomaly indicator, a one-standard-deviation increase in exposure ($SD = 0.19$) is associated with a 14.2% relative reduction from a post-2010 baseline of 12.3 per 1,000 births ($p < 0.10$). The effect is more precisely estimated for the pollution-linked index: a one-standard-deviation increase reduces its incidence by 25.7% relative to a post-2010 baseline of 4.02 per 1,000 births.

Table 3 disaggregates the pollution-linked index by diagnostic group. Cardiac malformations drive the largest effect in both absolute and relative terms: the coefficient of -0.004724^{**} implies a 39.5% relative reduction at one standard deviation of exposure, from a post-2010 baseline of 2.27 per 1,000 births. This finding is consistent with established biological pathways – fine particulate matter and combustion gases generate reactive oxygen species that cross the placental barrier and disrupt conotruncal septation. Neural tube defects contribute a smaller but significant reduction of 12.3% at one standard deviation ($p < 0.01$). Orofacial clefts show a small, statistically insignificant coefficient ($p = 0.34$),

¹⁰The LBW coefficient is, if anything, more precisely estimated with fewer controls, ruling out a particular control choice as the source, and is absent at the municipality-year aggregate level, consistent with the small individual-level estimate falling below what the aggregate specification's more limited sample size can resolve. For completeness, we also note that the technical change reduces the share of births to mothers with completed secondary education or more (coefficient = -0.034 , $p = 0.028$) – a compositional shift, plausibly linked to the labor-market restructuring documented by [Costa et al. \(2025\)](#) for the same setting, that offers one account of how a threshold-specific artifact like this could arise without any true deterioration in fetal health.

consistent with their primary teratogenic pathway running through organophosphate pesticide exposure rather than combustion-derived PM – and sugarcane cultivation is not pesticide-intensive in the relevant sense.

Finally, we find no socioeconomic gradient in the anomaly reduction. Repeating the results for sub-samples by maternal education group yields coefficients that are similar in magnitude and have fully overlapping standard errors across all groups. This suggests that the anomaly reduction reflects broad ambient PM exposure that benefits fetuses across the income distribution, rather than differential access to protective behaviors or healthcare.

4.3 Mortality

Table 4 presents results for respiratory and circulatory mortality, for the full population and for residents aged 60 and over, estimated by PPML with population as exposure (see the discussion of Table 5 below). Respiratory mortality shows no response: at one standard deviation of exposure ($SD = 0.19$), the point estimates imply changes of +0.4% for the full population and -0.3% among the elderly, and the 95% confidence interval for the full population excludes reductions larger than about 3%. Circulatory mortality coefficients are negative and significant at the 10% level, implying reductions of 2.5% for the full population and 3.0% among the elderly at one standard deviation. As with circulatory hospitalizations, however, circulatory mortality in high-sugarcane municipalities was already converging before the technical change: in an event study over 2001–2023, the pre-treatment coefficients are positive, jointly significant, and decline toward 2010. These circulatory estimates therefore cannot be attributed to the transition.

The null result is informative. Mortality is the least sensitive margin in the pollution–health dose–response relationship, and the particulate matter reductions achieved during the harvest season may fall below the threshold required to produce detectable effects on adult mortality. As we show next, the same holds for infants and children.

Table 5 restricts the mortality analysis to infant and child age groups. Because child deaths from respiratory causes are rare – most municipality-years record none – we estimate these regressions by Poisson pseudo-maximum likelihood (PPML), with the number of deaths as the outcome and the logarithm of live births as an offset (Santos Silva and Tenreiro, 2006). With this many zeros, the logarithm of a rate plus a small constant would make the estimates depend on the arbitrary choice of that constant (Chen and Roth, 2024). Infant respiratory mortality (Panel A) – deaths from respiratory causes (ICD

J00–J99) among children under five – shows no detectable response to the technical change: the point estimate implies a change of about -1% at one standard deviation of exposure ($SD = 0.19$), and the 95% confidence interval excludes reductions larger than about 6%. Results are similar in the pre-COVID sample and with the GDP control. All-cause infant mortality (Panel B) shows small positive coefficients, significant only at the 10% level in two columns, and respiratory mortality among children aged 5–11 years (Panel C) is indistinguishable from zero.

Together with the adult results, these estimates show that the reduction in particulate matter did not translate into detectable changes in mortality at any age. Mortality is the least sensitive margin of the pollution–health dose–response relationship: the health gains from eliminating pre-harvest burning appear in morbidity – hospitalizations and congenital anomalies – rather than in deaths.

5 The Pollution Channel

One natural concern is whether the health improvements documented above flow through the smoke-reduction channel rather than some confounding factor. Three patterns in the data support the smoke interpretation. Satellite data confirm that fire use on sugarcane fields fell sharply in municipalities more exposed to the technical change – and only during the harvest season, when burning would have occurred. Ground-level particulate matter concentrations fell during those same months, with no detectable effect in the off-season. And hospitalization reductions are disproportionately concentrated in harvest months, consistent with an acute smoke-exposure mechanism rather than a permanent shift in local health conditions.

To confirm that the technical change reduced burning specifically within sugarcane fields – rather than displacing fire to other land uses or seasons – we cross-reference satellite burned-area data with sugarcane land cover, computing burned area separately for the harvest season (July–September) and the off-season (January–March).¹¹

Virtually all sugarcane burning occurs during the harvest season. Off-season burned area remains essentially flat near zero throughout the entire period, providing a natural placebo that recurs in the particulate-matter results below. Panel A of Table 6 quantifies the harvest-season decline within municipalities, controlling for common temporal shocks

¹¹MODIS burned-area data and MapBiomass sugarcane land cover, processed via Google Earth Engine; see Section 3.1 for details.

through municipality and year fixed effects. The outcome is $\ln(\text{area burned in sugarcane pixels} + 0.001)$ during the harvest season. Municipalities more exposed to the technical change experienced a reduction of 4.96 log points per unit of exposure ($p < 0.01$, $SE = 0.45$). At the mean sugarcane land share ($\overline{Land} \approx 0.15$), this corresponds to approximately a 52.5% reduction in harvest-season burned area relative to the pre-mechanization baseline. This confirms that higher exposure translated into less burning exactly where and when the transition required.

Panel B of Table 6 turns from fire to air quality, showing that the burning reduction transmitted directly into lower PM concentrations. Using daily PM_{10} and $PM_{2.5}$ estimates from the CAMS global reanalysis, aggregated to municipality-season-year cells, we estimate equation (1) separately for harvest and off-season months.

At one standard deviation of municipal sugarcane land share ($SD = 0.19$), the technical change reduced PM_{10} by 1.8% and $PM_{2.5}$ by 1.8% during the harvest season (0.096×0.19 and 0.094×0.19 , respectively). The off-season coefficients are statistically indistinguishable from zero ($p > 0.4$ for PM_{10} ; $p > 0.8$ for $PM_{2.5}$): concentrations fall by only 0.5% and 0.1%, respectively, in January–March. This seasonal contrast is another evidence for the smoke mechanism. Panels A and B together establish the first two links: the technical change reduced actual fire use on sugarcane fields, and this translated into lower particulate matter concentrations during the months when burning occurs.

The third link connects reduced PM to reduced hospitalizations. Table 7 presents estimates of equation (2), which exploits monthly variation in hospitalizations to test whether health improvements concentrate in the burning season.

For respiratory hospitalizations, the year-round effect is -0.122^{***} , and the additional harvest-season increment is a highly significant -0.064^{***} : the respiratory reduction concentrates in July–September, exactly when burning would have occurred. For circulatory hospitalizations, the off-season baseline effect is -0.071^{***} , but the harvest-season increment is small and insignificant (-0.016), consistent with the circulatory channel being more diffuse and less directly linked to acute airway exposure. Satellite fire detection, ground-level PM monitoring, and monthly hospitalization records all point to the same conclusion: the health improvements documented here flow through the smoke-reduction channel. Results are robust to replacing the discrete harvest-season indicator with a continuous month-of-year pollution-intensity weight (Section 6.5), which yields an even more precisely estimated interaction coefficient.

6 Identification and Robustness

6.1 Parallel Trends and Event Study

We test the parallel trends assumption by estimating a dynamic version of equation (1):

$$\ln Y_{it} = \sum_{\tau \neq -1} \beta_{\tau} (D_{\tau} \times Land_i) + \theta_i + \gamma_t + \varepsilon_{it}$$

where D_{τ} is an indicator for year τ relative to 2011, the first treated year; 2010 ($\tau = -1$) is the reference year. Table 8 shows selected coefficients.

For respiratory hospitalizations, the 2008 coefficient is close to zero, and the only pre-treatment deviation is a positive 2009 coefficient that coincides with the H1N1 influenza pandemic and disappears once that year is excluded, leaving post-treatment estimates unchanged (Appendix H.1). For circulatory hospitalizations, the pre-treatment coefficients are positive and jointly significant ($p = 0.04$), declining toward 2010 whether or not 2009 is included. This pattern is consistent with circulatory outcomes in high-sugarcane municipalities already converging before the technical change, so the circulatory estimates cannot be distinguished from a continuation of that pre-existing convergence. Respiratory post-treatment coefficients grow steadily in magnitude over time, consistent with the gradual and accelerating elimination of burning under the GEP.

6.2 Wind Direction Triple-Difference

Table 9 presents results from equation (1) augmented with $WindExp_{it}$, which measures the directional exposure of municipality i to smoke from upwind sugarcane areas. This variable captures spillover effects from neighboring municipalities: reductions in burning in upwind neighbors reduce particulate exposure in downwind municipalities, independent of each municipality's own sugarcane exposure. Specifically, we estimate:

$$\ln(Y_{it}) = \alpha \cdot TC_{it} + \beta \cdot WindExp_{it} + \theta_i + \gamma_t + \epsilon_{it} \quad (5)$$

where α captures the direct effect of own technical change and β captures the additional exposure reduction from upwind neighbors' transitions. This is not an interaction—rather, two independent channels of health improvement operating through the smoke channel.

The TC coefficient remains negative and significant, reassuringly close to baseline, con-

firming that own-municipality mechanization reduces local health burdens. The *WindExp* coefficient is also negative—municipalities more downwind of sugarcane areas experience lower hospitalizations in years when upwind neighbors have reduced burning—and is significant at the 10% level for respiratory hospitalizations in the full sample only; the other three estimates are not statistically significant. For circulatory hospitalizations, the sign is consistent but imprecise. This pattern is suggestive rather than conclusive evidence that spillover effects from upwind neighbors operate through the airborne smoke channel transmitted by prevailing winds.

6.3 Placebo Outcomes

We assess whether the estimates reflect a general decline in hospitalizations unrelated to smoke by estimating the baseline specification for hospitalization groups with no plausible link to sugarcane smoke: injuries, falls, and transport accidents (ICD-10 S, V, W), fractures, poisoning and other external effects (T, excluding burns), appendicitis (K35–K37), and neoplasms (C00–C97), whose long latency rules out a response within a decade. Because several of these groups are small and record no hospitalizations in some municipality-years, we estimate the baseline specification by PPML, which accommodates zeros without an arbitrary constant (Chen and Roth, 2024). For each group c , we estimate:

$$E[H_{it}^c] = \exp(\alpha_i + \alpha_t + \beta^c \cdot TC_{it} + \ln \text{Pop}_{it})$$

where H_{it}^c is the number of hospitalizations in group c . Figure 1 compares these estimates with those for respiratory and circulatory hospitalizations and for all other causes combined. Three patterns stand out. First, every placebo group is statistically indistinguishable from zero, with point estimates between -0.094 and 0.024 . Pooling them into a single group of exogenous causes yields -0.027 ($SE = 0.051$), which rules out declines larger than about 0.13 log points. Second, the respiratory coefficient (-0.344) is more than ten times larger, and the difference is significant: estimating both groups jointly, the respiratory coefficient exceeds the placebo coefficient by -0.317 ($SE = 0.099$). Third, hospitalizations for all other causes, excluding respiratory and circulatory diseases, also decline (-0.109), by about a third of the respiratory effect. Part of this decline comes from infectious and parasitic diseases (ICD-10 chapter A) and is driven by intestinal infections (A00–A09), for which there is no plausible smoke pathway; excluding chapter A, the coefficient is -0.085 ($SE = 0.044$), about a quarter of the respiratory effect. Cleaner air is expected to improve health beyond the respiratory tract, and the resulting gradient—largest for respiratory

diseases, smaller for circulatory and other diseases, and nil for causes that smoke cannot affect—follows the ordering of biological plausibility.

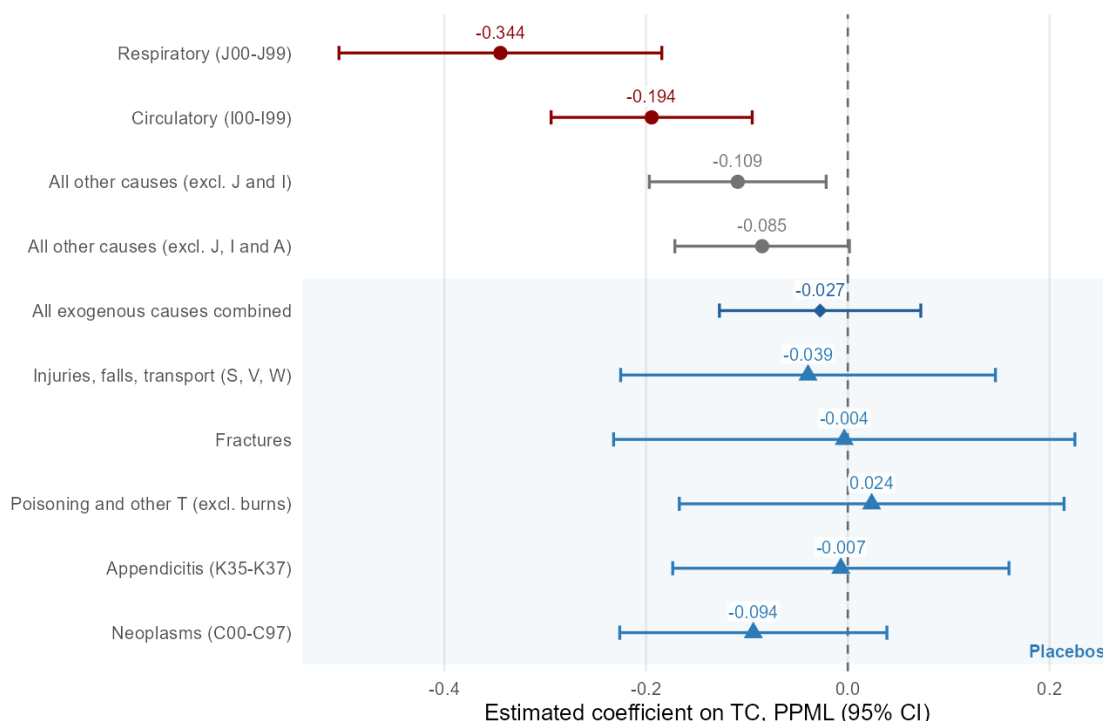


Figure 1: Respiratory Hospitalizations versus Placebo Outcomes

Notes: Each point is the coefficient on TC_{it} from a separate Poisson pseudo-maximum likelihood (PPML) regression of the number of hospitalizations in the indicated group, with municipality and year fixed effects and $\ln(\text{population})$ as offset, 2008–2021; bars are 95% confidence intervals with standard errors clustered by municipality. Hospitalizations are assigned to the patient’s municipality of residence. “All exogenous causes combined” pools injuries, falls and transport accidents (S, V, W), poisoning and other external effects (T, excluding burns T20–T32), appendicitis (K35–K37), and neoplasms (C00–C97); fractures are a subset of S. Respiratory and circulatory estimates differ slightly from Table 1, which uses $\ln(\text{hospitalizations per capita})$; PPML is used here because several placebo groups have municipality-years with no hospitalizations. Excluding also digestive diseases (K), the coefficient for all other causes is -0.069 ($SE = 0.047$); the decline in K is concentrated in elective surgeries (hernia, gallbladder) and shows pre-trends, inconsistent with a smoke channel.

We apply the same logic to congenital anomalies. The placebo is a group of anomalies with predominantly genetic or positional origins and no plausible link to combustion-derived particulate matter: polydactyly and syndactyly (ICD-10 Q69–Q70), hip, foot, and other positional musculoskeletal deformities (Q65–Q66, Q68), and minor anomalies of the ear, face, neck, and tongue (Q17–Q18, Q38). We exclude anomalies with documented environmental or chemical risk factors, such as abdominal wall defects, limb reduction defects, hypospadias, cryptorchidism, and chromosomal anomalies. Together, the placebo anomalies are about as common as the pollution-linked index (4.5 versus 4.0 per 1,000 births after 2010). Figure 2 shows that they do not respond to the technical change: at one

standard deviation of exposure, the placebo group changes by -6.0% (95% CI from -18.5% to $+6.4\%$), against -25.7% for the pollution-linked index, and the difference between the two is significant ($p < 0.001$). The contrast also speaks to measurement: differential changes in prenatal diagnosis or birth recording across municipalities would affect all anomalies recorded at birth, not only those with a pollution pathway.

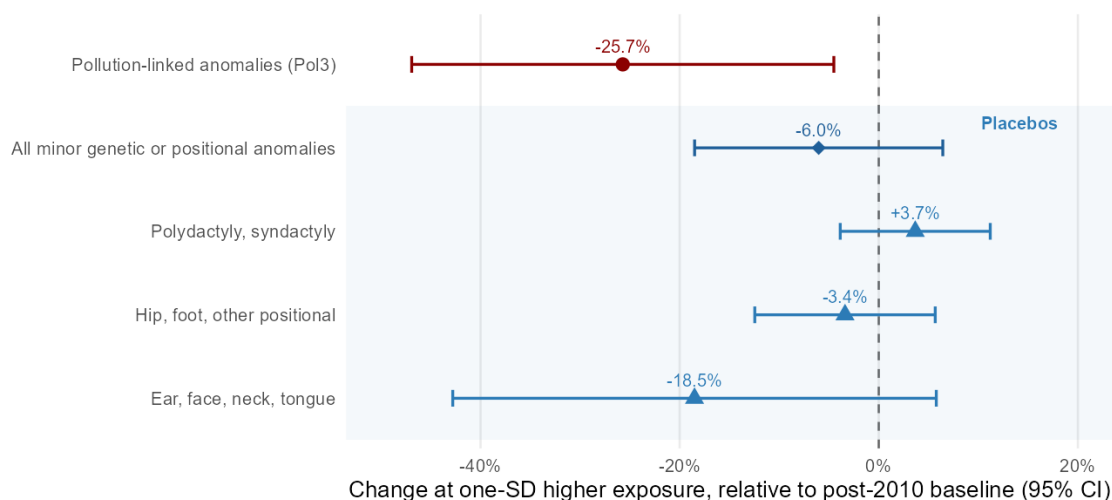


Figure 2: Pollution-Linked versus Placebo Congenital Anomalies

Notes: Each point is the coefficient on TC_{it} from a separate individual-level linear probability model (SINASC, São Paulo, 2002–2023; $N = 12,564,818$) with municipality, year, maternal education, sex, and marital-status fixed effects, expressed as the change at one standard deviation of exposure (0.19) relative to the post-2010 mean of each outcome; bars are 95% confidence intervals with standard errors clustered by municipality. Pol3 comprises neural tube defects (Q00–Q07), orofacial clefts (Q35–Q37), and cardiac malformations (Q20–Q26). The placebo group comprises Q65–Q66, Q68–Q70, Q17–Q18, and Q38. The difference between the Pol3 and placebo coefficients, estimated in a single regression of their difference, is -0.0040 (SE = 0.0011).

Figure A-16 in the Appendix extends the exercise to every ICD-10 chapter, using $\ln(\text{hospitalizations per capita} + 0.001)$. Two patterns stand out. First, respiratory diseases (J) show one of the largest negative effects. Second, the placebo groups – injuries and trauma (S) and falls and drowning (W) – are statistically indistinguishable from zero, validating the placebo choice; transport accidents (V) are rarely recorded as the principal diagnosis and cannot be estimated separately.

The majority of the remaining groups are also indistinguishable from zero. Four show significant negative coefficients – infectious and parasitic (A), blood and immune (D), digestive (K), and renal and urinary (N) – a pattern consistent with the broader systemic effects of $PM_{2.5}$ documented in the literature, as fine particles trigger inflammatory cascades that extend beyond the respiratory tract (Organization, 2021). Establishing causality for each of these groups is beyond the scope of this paper. The infectious-disease coefficient is

as large as the respiratory one, which may reflect systemic effects of fine particulate matter but also broader changes in hospital use in high-sugarcane municipalities that we cannot rule out. Crucially, the overall pattern does not support a confounding interpretation: any confounder driving the respiratory and circulatory results would need to leave trauma, mental health, neoplasms, pregnancy, and most other diagnostic groups entirely unaffected – a highly implausible requirement.

6.4 Continuous Treatment Intensity

The binary specification assumes full treatment from 2011 onward; in reality, the GEP phased in gradually. We replace $D_{t>2010}$ with $f_{aset} \in [0, 1]$, defined as the fraction of pre-2011 baseline sugarcane burning emissions already eliminated by year t (from SEEG data (SEEG, 2024)):

$$f_{aset} = \frac{\text{Baseline}_{SP} - \text{Emiss}_{SP,t}}{\text{Baseline}_{SP}}$$

This variable is common to all municipalities in a given year, providing an exogenous state-level shifter that interacts with the municipal-level sugarcane land share. Table 10 presents results alongside the binary estimates for comparison.

The continuous treatment estimates are larger in magnitude— -0.576 for respiratory and -0.218 for circulatory hospitalizations—but at the mean value of $f_{aset} \approx 0.56$ imply effects nearly identical to the binary estimates (-0.323 vs. -0.330 for respiratory). This confirms that the binary specification captures the average effect correctly, and that the health improvements build up proportionally with the actual pace of the emissions reduction.

6.5 Continuous Harvest-Season Intensity

The harvest-season triple-difference in equation (2) uses a binary indicator, $\text{Harvest}_m = \mathbf{1}(m \in \{7, 8, 9\})$, that treats all three harvest months as equivalent and all nine off-season months as equivalent. As a robustness check, we replace this binary indicator with a continuous, month-specific intensity weight built from the pre-2011 seasonal profile of particulate matter. For each pollutant $p \in \{PM_{10}, PM_{2.5}\}$, we compute the mean concentration by calendar month, pooled across all municipalities and days in 2003–2010 (i.e., before the technical change began, to avoid circularity):

$$\overline{PM}_m^p = \frac{1}{N_m} \sum_i \sum_{t \in [2003, 2010]} \sum_{d \in m} PM_{i,t,d}^p, \quad m = 1, \dots, 12$$

normalized to sum to one across months, $HarvestShare_m^p = \overline{PM}_m^p / \sum_{m'} \overline{PM}_{m'}^p$, and averaged across the two pollutants to obtain $HarvestShare_m$. This weight is fixed across municipalities and years, varying only by calendar month. We then re-estimate equation (2), replacing $Harvest_m$ with $HarvestShare_m$:

$$\ln Y_{itm} = \alpha \cdot TC_{it} + \beta \cdot TC_{it} \times HarvestShare_m + \theta_{i \times m} + \gamma_t + \varepsilon_{itm}$$

Table 11 compares the two specifications.

The continuous specification confirms the harvest-season pattern documented above: the interaction coefficient is estimated with substantially more precision for respiratory hospitalizations ($p < 0.001$) and is significant at the 5% level for circulatory hospitalizations ($p = 0.014$), where the binary specification found no detectable effect. Because $HarvestShare_m$ is constructed independently of the treatment – from the pre-2011 pollution calendar, not from realized post-treatment burning – this result is not mechanically guaranteed, and provides additional evidence that health improvements track the intensity of historical burning activity within the year, not just a discrete in-season/off-season distinction.

6.6 Excluding the City of São Paulo

São Paulo city, the state capital, has a population of approximately 12 million but essentially zero sugarcane land share, placing it permanently in the control group. Its size means it can dominate population-weighted regressions. Table 12 shows that results are robust to its exclusion.

Coefficients remain negative and statistically significant after removing São Paulo city; the respiratory estimate is, if anything, slightly larger, and the circulatory estimate is essentially unchanged.

6.7 Alternative Pre-Treatment Averaging Windows

Our baseline treatment variable averages each municipality’s sugarcane land share over 2000–2010. To verify that this particular window is not driving the results, we recompute the treatment variable using two shorter windows immediately preceding the 2011 cutoff – 2005–2010 and 2008–2010 – and re-estimate equation (1) with each. Table A-2 shows that estimates are stable across windows: the implied effect of a one-standard-deviation increase

in exposure ranges from -6.9% to -7.1% for respiratory hospitalizations and from -3.2% to -3.4% for circulatory hospitalizations, with all coefficients significant at the 1% level throughout. The choice of averaging window is therefore immaterial to our conclusions.

Our treatment assigns exposure to every municipality with a positive sugarcane land share. A natural alternative would treat municipalities with very small shares as unexposed. Table A-3 shows that this choice is immaterial: imposing minimum-share cutoffs between 1% and 15% moves the respiratory coefficient between -0.387 and -0.357 and the circulatory coefficient between -0.179 and -0.151 , all significant at the 1% level. The reason is that the 204 municipalities with positive shares at or below 5% – nearly a third of the sample – account for less than 3% of the state’s total sugarcane share.

A related concern is whether the linear dose-response imposed by the continuous treatment is appropriate (Callaway et al., forthcoming). Table A-4 replaces TC_{it} with post-2010 indicators for quartiles of sugarcane land share among the 553 municipalities with positive share, using the 89 municipalities without sugarcane as the reference group. The response rises with exposure: the two lowest quartiles, with mean shares of 0.5% and 5.5%, show no significant effect on respiratory hospitalizations (0.002 and -0.034), while the two upper quartiles show reductions of -0.138 and -0.171 , both significant at the 1% level. Circulatory hospitalizations follow the same pattern, with smaller magnitudes, and terciles and quintiles of sugarcane share yield the same pattern.

7 Discussion

7.1 Cost Savings from Mechanization

Lower hospitalization rates and reduced congenital anomaly incidence represent direct cost savings to the Brazilian public health system. To quantify these savings conservatively, we multiply the estimated reduction in each outcome by the average 2019 SUS reimbursement cost for that hospitalization category in São Paulo, using a single reference year to avoid inflation-adjustment assumptions.¹²

For respiratory hospitalizations, a one-standard-deviation increase in exposure reduces hospitalizations by 7.3%. Aggregating the implied reduction across all exposed municipalities over 2011–2021 yields approximately 74,000 fewer respiratory admissions. At an

¹²All figures are expressed in 2019 Brazilian reals. In 2019, the average BRL/USD exchange rate was approximately R\$ 3.94 per dollar (PTAX/BCB annual average).

average 2019 SUS reimbursement cost of R\$ 1,134 per admission, this translates into R\$ 84 million (USD 21 million) in avoided hospitalization expenditures over the decade. We do not add circulatory hospitalizations to this total, because the circulatory estimates cannot be distinguished from a pre-existing convergence (Section 6.1). Taken at face value, they would imply approximately 37,000 fewer circulatory admissions and, at an average 2019 cost of R\$ 2,739 per admission, an additional R\$ 100 million (USD 25 million), which we treat as an upper bound rather than as part of our estimate.

The mechanization also reduced the incidence of congenital cardiac malformations (ICD-10 Q20–Q26), the anomaly type with the strongest point estimate in our analysis. At the estimated coefficient of -0.004724 per birth (Table 3), we estimate approximately 2,504 fewer births with cardiac malformations across exposed municipalities over 2011–2023. Each such birth generates at minimum one surgical hospitalization: at an average 2019 SUS reimbursement of R\$ 13,622 for cardiac malformation admissions, this lower bound corresponds to R\$ 34 million (USD 9 million) in avoided costs.

Together, the avoided respiratory hospitalization and cardiac anomaly costs amount to R\$ 118 million (USD 30 million) in 2019 reais. This figure is a very conservative lower bound. It excludes outpatient visits, pharmaceuticals, productivity losses, and the costs associated with neural tube defects and orofacial clefts – the other two components of the Pol3 index. It also excludes the BPC/LOAS social protection transfer (approximately one minimum wage per month, lifetime) that accrues to children with permanent disability resulting from severe cardiac malformations.

8 Final Remarks

Burning is cheap for growers and costly for everyone around. We found that banning agricultural fires at scale in Brazil’s sugarcane sector improved the health of surrounding populations. This improvement does not seem to have occurred by chance: every piece of evidence we present points to the same causal chain – less burning, less particulate matter, better health. The effect is also stronger during the harvest season, exactly when burning would otherwise have occurred, and weaker or absent outside it. To our knowledge, this is among the largest-scale eliminations of agricultural fire studied to date.

The pattern of effects across outcomes is itself informative about the biological mechanisms at work. Mortality is the least sensitive margin in the pollution–health dose–response relationship. Hence, the PM reductions achieved during the harvest season fall below the

threshold required to produce detectable effects on mortality – among adults and among infants and children alike. The congenital anomaly results also reveal developmental damage during organogenesis that leaves no trace in birthweight yet reduces cardiac malformations and neural tube defects. This hierarchy – null mortality at all ages, significant hospitalizations, significant anomalies – reflects a simple idea: the same pollution shock affects different health margins at different thresholds, with morbidity and fetal development responding at exposure levels below those required to raise mortality.

Two of the outcomes we document have not previously been examined in the sugarcane burning literature. Cardiac malformations and neural tube defects have established biological pathways to combustion-derived particulate matter ([Tanwar et al., 2017](#); [Chen et al., 2024](#)), yet no prior study has linked sugarcane burning to these outcomes with causal identification.

There is likely some avoidance behavior, in the sense that residents of cane-intensive municipalities may partially offset smoke exposure through air conditioning, keeping windows closed, or reducing outdoor activity during the harvest season. This welfare reduction is not something we can measure here. There is also a potential confounder due to residential sorting: individuals who are more sensitive to air pollution may be disproportionately likely to avoid or leave municipalities with intensive sugarcane production, leaving behind a population selected toward lower sensitivity. We do not consider this latter mechanism to be particularly plausible because São Paulo is precisely the richest region in Brazil, and the dominant migration pattern is toward São Paulo rather than away from it.

Health improvements documented here carry direct policy implications wherever agricultural burning persists. Within Brazil, that means the northeastern states, which have not yet completed the transition – Pernambuco alone still cuts over 90% of its cane by hand ([Conab, 2024](#)). The GEP shows what closing that gap can look like: a voluntary agreement that traded sustainability credentials for preferential access to demanding export markets, aligning mill incentives with early adoption of cleaner technology. Extending that model to the Northeast, where labor market conditions and export exposure differ, is a direct and quantifiable policy opportunity – and the same logic applies internationally, wherever domestic regulation or supply-chain standards can price in the health externalities we document here.

Mechanization also displaced a substantial share of the manual cane-cutting workforce, concentrated among low-skilled labor – a cost this paper does not measure, and we do

not know whether the health gains documented here outweigh it. That is a distributional question for future work. What we can say is that eliminating pre-harvest burning left the people living near cane fields measurably healthier.

Table 1: Hospitalizations Primarily Caused by Respiratory and Circulatory Diseases

	$y = \ln(\text{hospitalizations per capita})$			
	(1) Full	(2) Full+GDP	(3) Pre-Covid	(4) Pre-Covid+GDP
<i>Panel A — Respiratory Diseases (ICD J00–J99)</i>				
Tech. Change (α)	−0.386*** (0.093)	−0.384*** (0.093)	−0.300*** (0.092)	−0.302*** (0.092)
ln(GDP)		−0.035 (0.065)		0.064 (0.051)
Observations	8,988	8,988	7,704	7,704
<i>Panel B — Circulatory Diseases (ICD I00–I99)</i>				
Tech. Change (α)	−0.178*** (0.055)	−0.182*** (0.054)	−0.177*** (0.050)	−0.179*** (0.050)
ln(GDP)		0.047 (0.042)		0.035 (0.045)
Observations	8,988	8,988	7,704	7,704
Municipality FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Weights	Pop.	Pop.	Pop.	Pop.
Period	2008–2021	2008–2021	2008–2019	2008–2019

Standard errors clustered by municipality in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable is the logarithm of hospitalizations per capita primarily caused by respiratory diseases (Panel A) or circulatory diseases (Panel B), with hospitalizations assigned to the patient's municipality of residence. Pre-Covid columns exclude observations after 2019.

Table 2: Birth Outcomes — Individual-Level Linear Probability Model

	Congenital Anomalies		Birthweight	
	(1) Any Anomaly (SINASC flag)	(2) Pol3 Index (NTD + Orofac. + Card.)	(3) LBW (<2,500g)	(4) VLBW (<1,500g)
Tech. Change (α)	−0.00922* (0.00486)	−0.00545** (0.00229)	0.00333** (0.00168)	−0.00033 (0.00073)
Observations	12,009,629	12,564,818	12,564,818	12,564,818
Municipality FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Education FE	✓	✓	✓	✓
Sex FE	✓	✓	✓	✓
Marital status FE	✓	✓	✓	✓
Baseline mean	0.01231 ^a	0.00402 ^a	0.09209	0.01435

Standard errors clustered by municipality in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Individual SINASC birth records for São Paulo, 2002–2023. Column (1): any congenital anomaly, as recorded in the SINASC anomaly field (births with the field missing or marked as ignored are excluded, hence the smaller N). Column (2): Pol3 index, equal to one if the birth presents at least one of: neural tube defects (Q00–Q07), orofacial clefts (Q35–Q37), or congenital cardiac malformations (Q20–Q26). Columns (3)–(4): birthweight below 2,500 g and 1,500 g. $TC_{it} = Land_i \times D_{t>2010}$.

^a Computed over 2011–2023 rather than the full sample, since recorded anomaly rates rise substantially over 2002–2023 as prenatal ultrasound and fetal echocardiography access expanded; a full-sample baseline would understate current rates. Coefficients are unaffected, as year fixed effects absorb this common trend.

Table 3: Congenital Anomalies by Diagnostic Group

$y = \text{Pr}(\text{anomaly type})$	(1) Neural Tube (Q00–Q07)	(2) Orofacial (Q35–Q37)	(3) Cardiac (Q20–Q26)	(4) Pol3 Index
Tech. Change (α)	−0.000743*** (0.000254)	−0.000122 (0.000128)	−0.004724** (0.002038)	−0.005445** (0.002291)
Observations	12,564,818			
Municipality FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Education FE	✓	✓	✓	✓
Sex FE	✓	✓	✓	✓
Marital status FE	✓	✓	✓	✓
Baseline, post-2010 (per 1,000)	1.15	0.76	2.27	4.02

Standard errors clustered by municipality in parentheses. ** $p < 0.05$, *** $p < 0.01$. Individual SINASC birth records for São Paulo, 2002–2023. Each column is a separate LPM regression with municipality, year, education, sex, and marital status fixed effects. Baseline rates are computed over 2011–2023 rather than the full sample: recorded anomaly rates rise substantially over 2002–2023 as prenatal ultrasound and fetal echocardiography access expanded, and a full-sample baseline would understate current rates and overstate relative effects. Coefficients are unaffected, as year fixed effects absorb this common trend.

Table 4: Mortality Primarily Caused by Respiratory and Circulatory Diseases

	Respiratory (J00–J99)		Circulatory (I00–I99)	
	All ages	Aged 60+	All ages	Aged 60+
	(1)	(2)	(3)	(4)
<i>Panel A — 2001–2023</i>				
Tech. Change (α)	0.021 (0.101)	−0.014 (0.119)	−0.134* (0.081)	−0.162* (0.089)
<i>Panel B — 2002–2021, controlling for $\ln(\text{GDP per capita})$</i>				
Tech. Change (α)	−0.010 (0.085)	−0.036 (0.101)	−0.132* (0.072)	−0.156** (0.079)
Municipality FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Offset	Pop.	Pop.	Pop.	Pop.

Standard errors clustered by municipality in parentheses. * $p < 0.1$, ** $p < 0.05$. PPML estimates, as in Table 5: the outcome is the number of deaths from the indicated cause in the municipality-year (SIM, by municipality of residence), with $\ln(\text{population})$ as offset, so that coefficients measure proportional changes in the mortality rate. In an event study over 2001–2023, pre-treatment coefficients for circulatory mortality are positive, jointly significant ($p < 0.01$), and decline toward 2010; circulatory estimates should be read with this caveat.

Table 5: Infant and Child Mortality by Cause and Age Group

	PPML: y = number of deaths, offset $\ln(\text{live births})$			
	(1)	(2)	(3)	(4)
	2001–2023	2001–2019	2002–2021 +GDP	2002–2019 +GDP
<i>Panel A — Infant respiratory mortality (J00–J99, <5 years)</i>				
Tech. Change (α)	–0.061 (0.138)	–0.016 (0.145)	–0.059 (0.150)	–0.040 (0.158)
<i>Panel B — Infant all-cause mortality (<5 years)</i>				
Tech. Change (α)	0.113* (0.060)	0.102 (0.065)	0.107* (0.062)	0.101 (0.066)
<i>Panel C — Children 5–11 years, respiratory mortality</i>				
Tech. Change (α)	–0.047 (0.427)	–0.000 (0.430)	–0.131 (0.455)	–0.033 (0.455)
Municipality FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Offset	Births	Births	Births	Births

Standard errors clustered by municipality in parentheses. * $p < 0.1$. Poisson pseudo-maximum likelihood (PPML) estimates. Outcome: number of cause-specific deaths in the municipality-year, with $\ln(\text{live births})$ as offset, so that coefficients measure proportional changes in the mortality rate per live birth. PPML accommodates the many municipality-years with zero deaths without an arbitrary constant inside the logarithm; municipalities with no deaths over the whole period carry no information under municipality fixed effects and drop out. Columns (3)–(4) add $\ln(\text{GDP per capita})$, available for 2002–2021. Sample: São Paulo municipalities, 2001–2023 (full) and 2001–2019 (pre-Covid).

Table 6: The Pollution Channel: Sugarcane Burning and Particulate Matter

	Harvest (Jul-Sep)	Off-season (Jan-Mar)
<i>Panel A: Sugarcane Burning (MODIS $\times$ MapBiomass)</i>		
$\ln(\text{Burned area in sugarcane, ha})$	-4.964^{***} (0.450)	—
Observations	15,216	
<i>Panel B: Particulate Matter (CAMS reanalysis)</i>		
$\ln(\text{PM}_{10})$	-0.096^{***} (0.010)	-0.005 (0.007)
$\ln(\text{PM}_{25})$	-0.094^{***} (0.010)	-0.001 (0.007)
Observations	14,124	14,124
Municipality FE	✓	✓
Year FE	✓	✓

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Standard errors clustered at the municipality level. Panel A: outcome is $\ln(\text{area burned in sugarcane pixels} + 0.001)$, harvest season (July–September). Off-season column omitted: the $\log(\cdot + 0.001)$ specification is unreliable when zeros dominate, as they do in January–March. Source: MODIS MCD64A1 burned-area product crossed with MapBiomass LULC sugarcane layer (Collection 9), processed in Google Earth Engine. Panel B: daily PM estimates from the CAMS global reanalysis (Saldanha et al., 2026b,a), aggregated to municipality-season-year, 2003–2024.

Table 7: Harvest Season Triple-Difference for Hospitalizations

$y = \ln(\text{hospitalizations/pop})$	Respiratory	Circulatory
TC (off-season baseline)	-0.122^{***} (0.030)	-0.071^{***} (0.019)
$TC \times \text{Harvest (Jul-Sep)}$	-0.064^{***} (0.009)	-0.016 (0.010)
Municipality $\times$ Month FE	✓	✓
Year FE	✓	✓
Weights	Pop.	Pop.

** $p < 0.05$, *** $p < 0.01$. Municipality $\times$ month fixed effects absorb local seasonal patterns in hospitalization rates. TC captures the year-round effect; $TC \times \text{Harvest}$ captures the additional effect during burning months (July–September).

Table 8: Event Study Coefficients — Selected Years

Year (τ relative to 2011)	Respiratory	Circulatory
$\tau = -3$ (2008)	+0.039 (0.058)	+0.120** (0.052)
$\tau = -2$ (2009)	+0.112*** (0.035)	+0.092** (0.042)
$\tau = -1$ (2010)	0 (reference)	
$\tau = 0$ (2011)	-0.128*** (0.044)	-0.081*** (0.028)
$\tau = 5$ (2016)	-0.311*** (0.103)	-0.077 (0.062)
$\tau = 10$ (2021)	-0.731*** (0.095)	-0.172* (0.099)

Standard errors clustered by municipality in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Respiratory: the 2008 coefficient is close to zero, while the 2009 coefficient is positive and significant, coinciding with the H1N1 influenza pandemic; excluding 2009 removes this pre-treatment deviation and leaves the post-treatment estimates unchanged (Appendix H.1). Circulatory: the pre-treatment coefficients are positive, jointly significant ($p = 0.04$), and decline toward the reference year, also when 2009 is excluded; circulatory estimates should be read with this caveat. Full event study figures are shown in Figures A-11 and A-12 in the Appendix.

Table 9: Wind Direction Triple-Difference

	Respiratory		Circulatory	
	Full	Pre-Covid	Full	Pre-Covid
TC	-0.329*** (0.077)	-0.257*** (0.076)	-0.158*** (0.048)	-0.156*** (0.043)
$WindExp$	-0.043* (0.026)	-0.031 (0.023)	-0.031 (0.022)	-0.015 (0.021)
Municipality FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓

* $p < 0.1$, *** $p < 0.01$. Clustered SEs (municipality). $WindExp_{it}$ is the directional exposure to upwind sugarcane burning, constructed from NASA POWER/MERRA-2 wind data (July–September). Figure A-14 maps $WindExp$ across municipalities.

Table 10: Continuous Treatment Intensity

	Respiratory	Circulatory
Binary TC (baseline)	-0.330^{***} (0.077)	-0.159^{***} (0.047)
Continuous TC ($\times f_{ase_t}$)	-0.576^{***} (0.152)	-0.218^{**} (0.090)
Municipality FE	✓	✓
Year FE	✓	✓
Weights	Pop.	Pop.

** $p < 0.05$, *** $p < 0.01$. The binary and continuous specifications are estimated separately. Implied effect at mean $f_{ase_t} \approx 0.56$: $-0.576 \times 0.56 \approx -0.323$, nearly identical to the binary estimate of -0.330 . Figure A-15 plots f_{ase_t} over time.

Table 11: Continuous Harvest-Season Intensity

$y = \ln(\text{hospitalizations/pop})$	Respiratory	Circulatory
Binary $TC \times \text{Harvest}$ (Jul–Sep)	-0.064^{***} (0.009)	-0.016 (0.010)
Continuous $TC \times \text{HarvestShare}_m$	-2.053^{***} (0.227)	-0.624^{**} (0.253)
Municipality $\times$ Month FE	✓	✓
Year FE	✓	✓
Weights	Pop.	Pop.

** $p < 0.05$, *** $p < 0.01$. HarvestShare_m ranges from 0.059 (December, lowest) to 0.127 (September, peak). The implied gap in effect between the peak and trough months is $\beta \times (0.127 - 0.059) = -0.140$ for respiratory and -0.042 for circulatory – larger than the binary estimate for respiratory, and directionally consistent for circulatory.

Table 12: Robustness: Excluding the City of São Paulo

	Respiratory		Circulatory	
	Full	Pre-Covid	Full	Pre-Covid
Baseline	−0.386***	−0.300***	−0.178***	−0.177***
Excl. SP city	−0.416***	−0.335***	−0.179***	−0.159***

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Each row reports the coefficient on TC from a separate regression; the excluded municipality is IBGE code 3550308.

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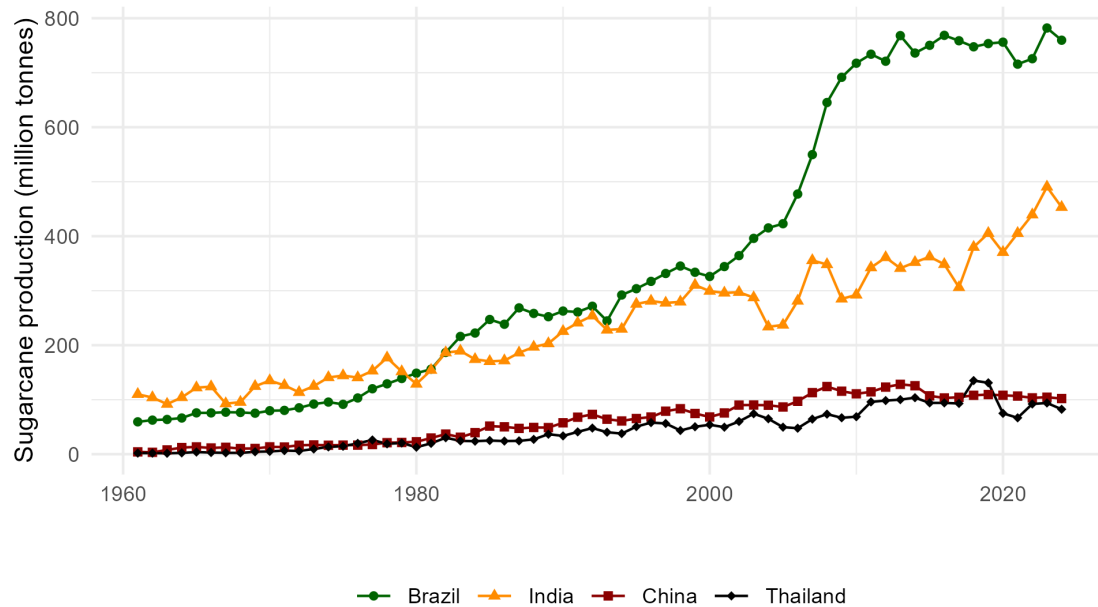
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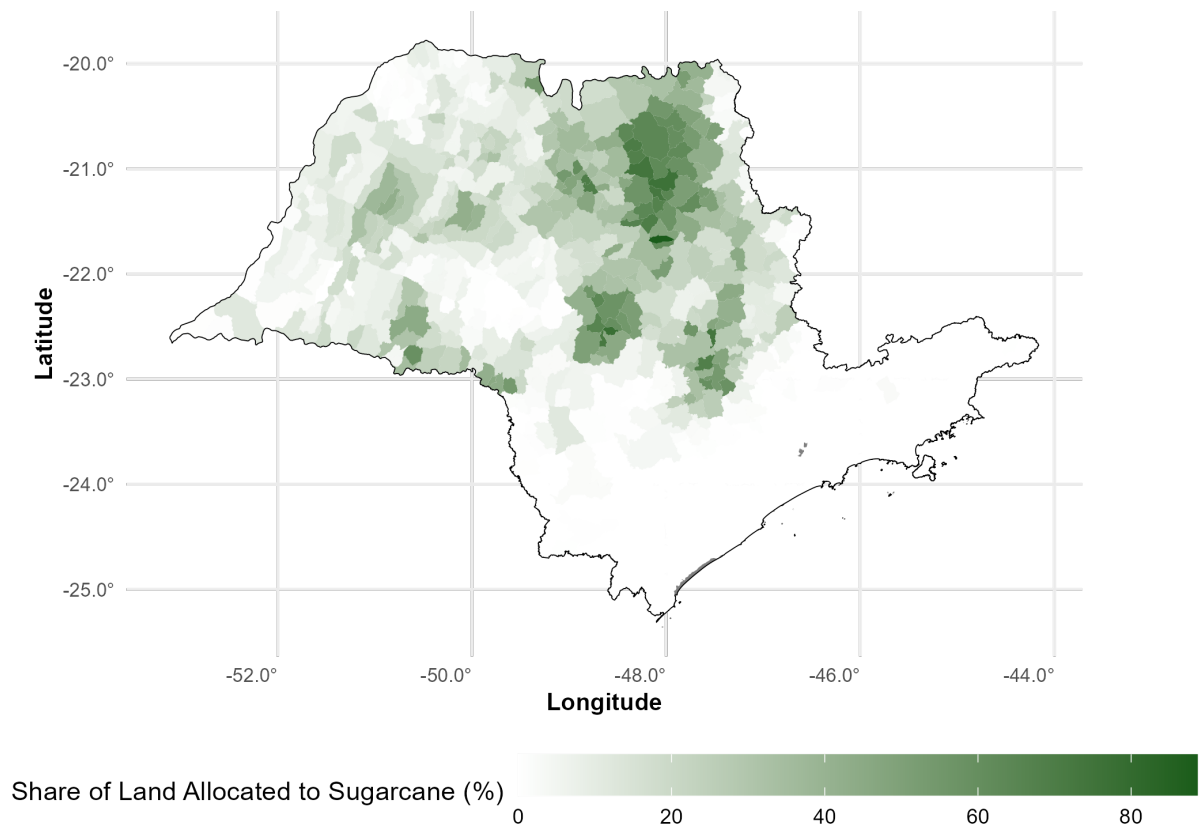
Appendix

Figure A-1: Sugarcane Production Trends for the Top 4 Producing Countries



Author's elaboration using data from [FAO \(2025\)](#), with major processing by Our World in Data. The figure displays annual sugarcane production in million tonnes for the four largest producing countries.

Figure A-2: Spatial Distribution of Land Allocated to Sugarcane Cultivation in São Paulo (2000–2010)



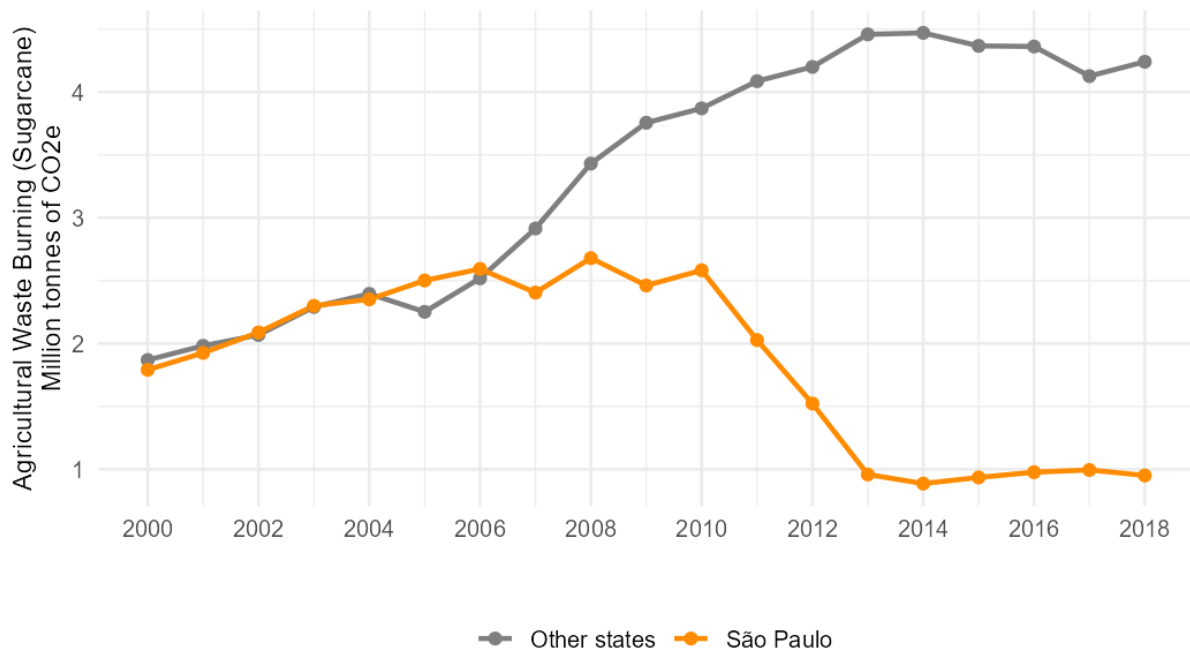
Data from IBGE-Sidra, Pesquisa Agrícola Municipal ([IBGE, 2024](#)). The figure shows the share of land allocated to sugarcane cultivation in each municipality between 2000 and 2010.

Figure A-3: Land Permitted for Pre-Harvest Burning in São Paulo



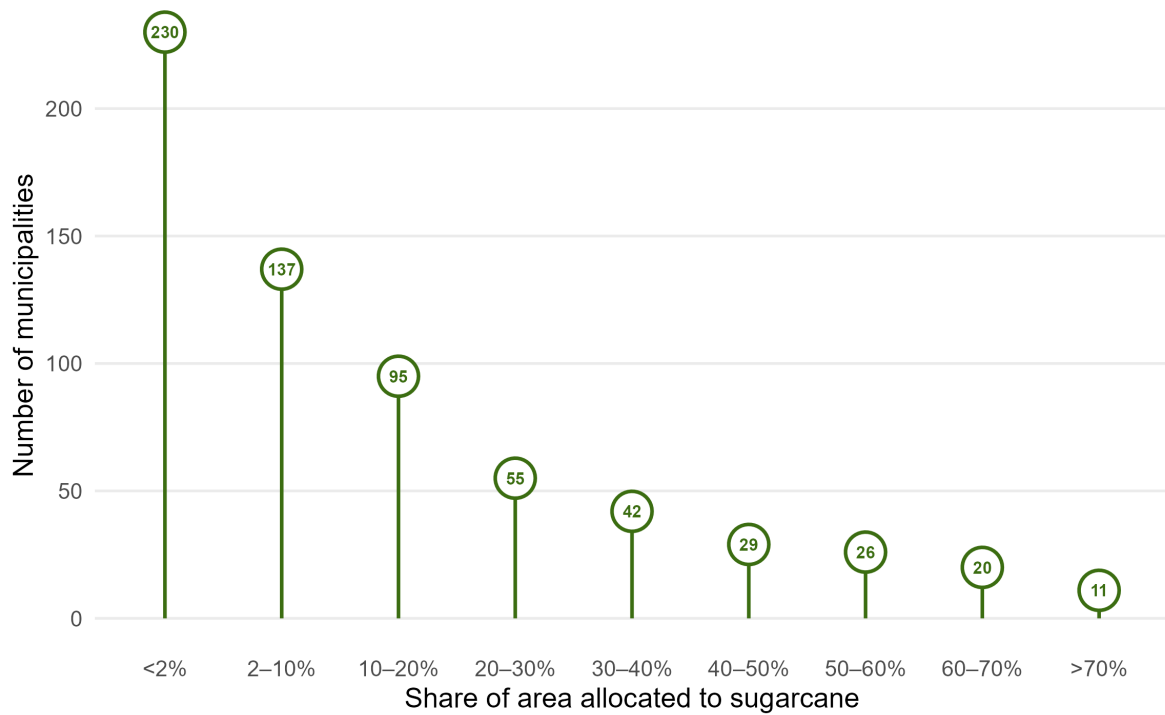
Data from [SAA-SP \(2023\)](#). The graph illustrates total area permitted for pre-harvest sugarcane burning. The gray dashed line marks when the significant reduction began.

Figure A-4: Sugarcane Burning Emissions in São Paulo State (SEEG)



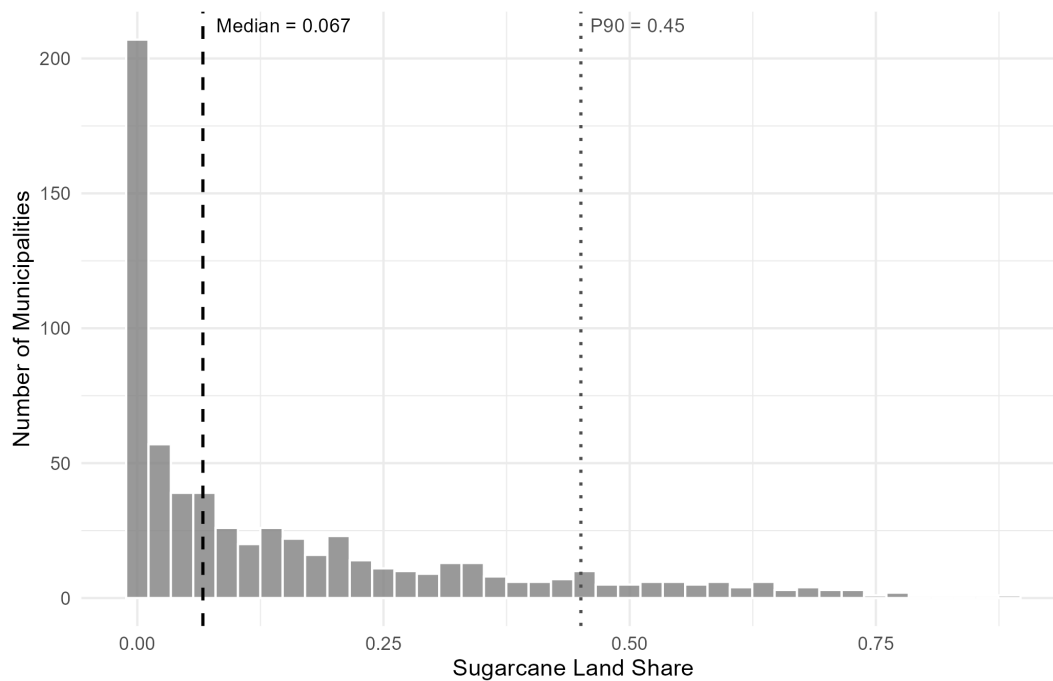
Annual sugarcane agricultural waste burning emissions (million tonnes of CO₂e). Orange line: São Paulo state. Gray line: all other Brazilian states combined. Source: SEEG (Sistema de Estimativas de Emissões e Remoções de Gases de Efeito Estufa), *Queima de Resíduos Agrícolas* series. São Paulo emissions declined sharply after 2010, coinciding with the phased implementation of the Green Ethanol Protocol (GEP, 2007), which mandated the progressive elimination of pre-harvest burning. Other states show no comparable trend, providing a natural comparison group.

Figure A-5: Distribution of Municipalities by Share of Agricultural Land Allocated to Sugarcane in São Paulo (2000–2010)



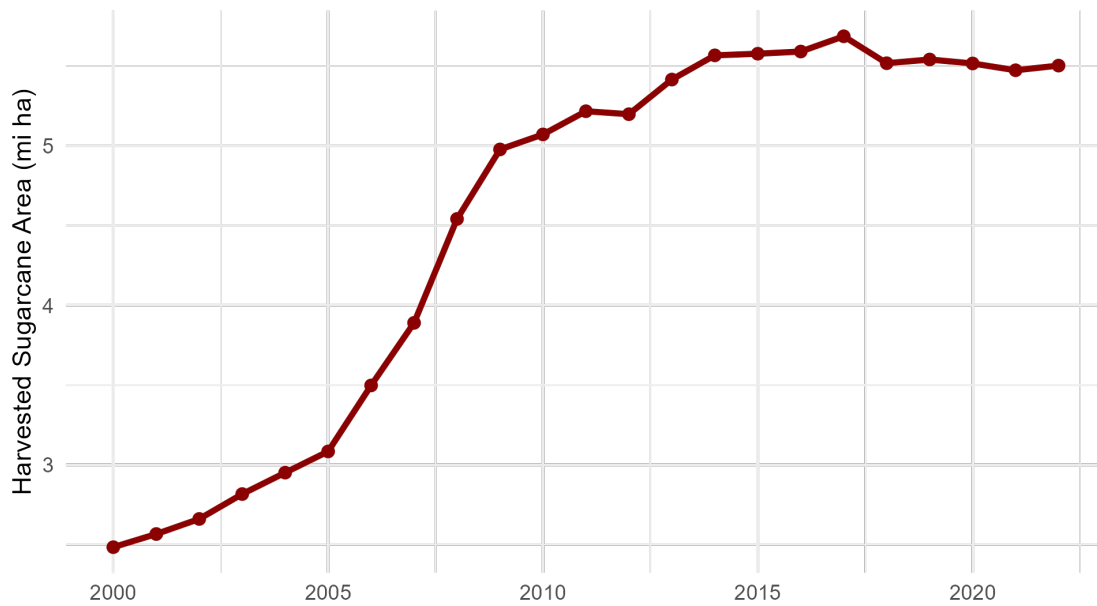
Data from IBGE-Sidra, Pesquisa Agrícola Municipal ([IBGE, 2024](#)). Each bin shows the number of municipalities whose share of total agricultural area devoted to sugarcane falls within the indicated interval. Shares are computed as the ratio of harvested sugarcane area to total harvested cropland area at the municipality level. The last category (>70%) pools all municipalities above that threshold.

Figure A-6: Distribution of Municipalities by Share of Agricultural Land Allocated to Sugarcane in São Paulo (2000–2010)



Data from IBGE-Sidra, Pesquisa Agrícola Municipal ([IBGE, 2024](#)). Shares are computed as the ratio of harvested sugarcane area to total harvested cropland area at the municipality level.

Figure A-7: Harvested Sugarcane Area in São Paulo



Data from IBGE-Sidra, Pesquisa Agrícola Municipal (IBGE, 2024). The figure depicts the harvested sugarcane area in São Paulo state, measured in million hectares.

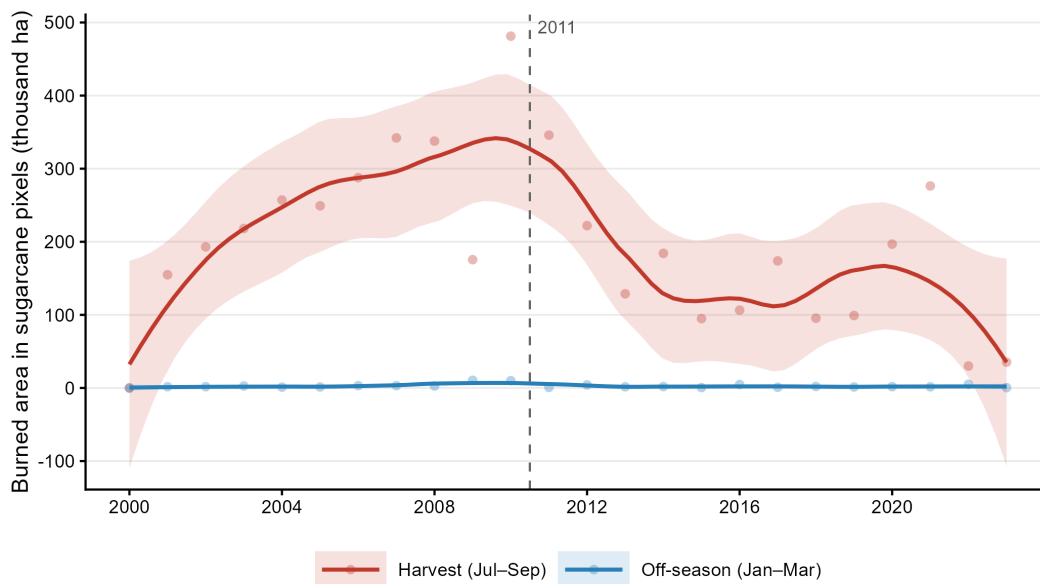
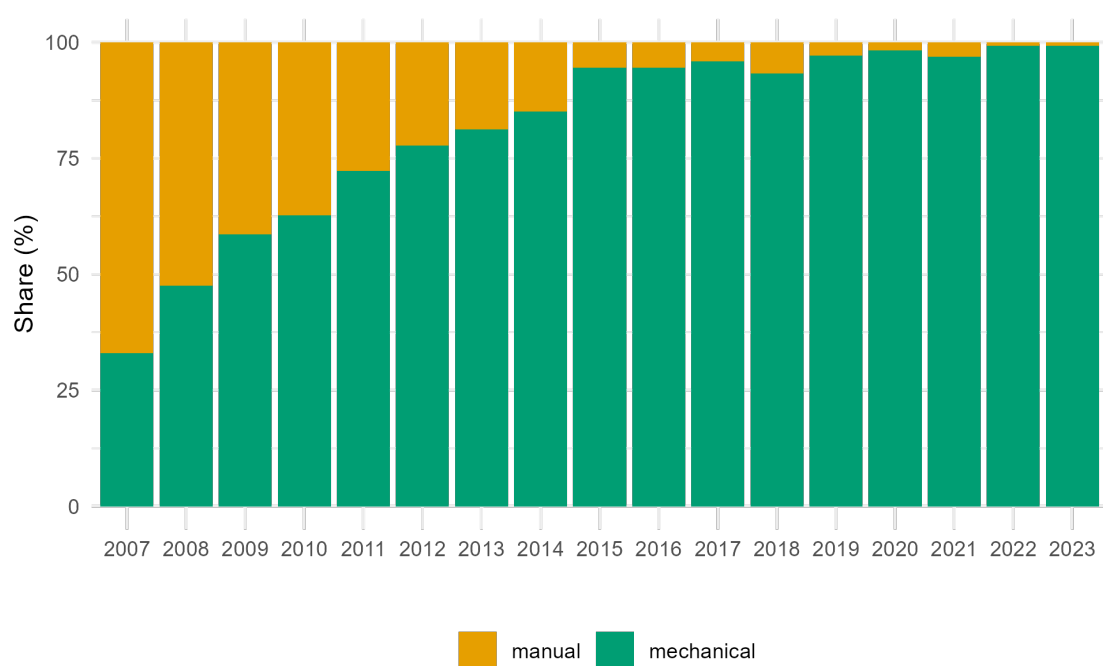


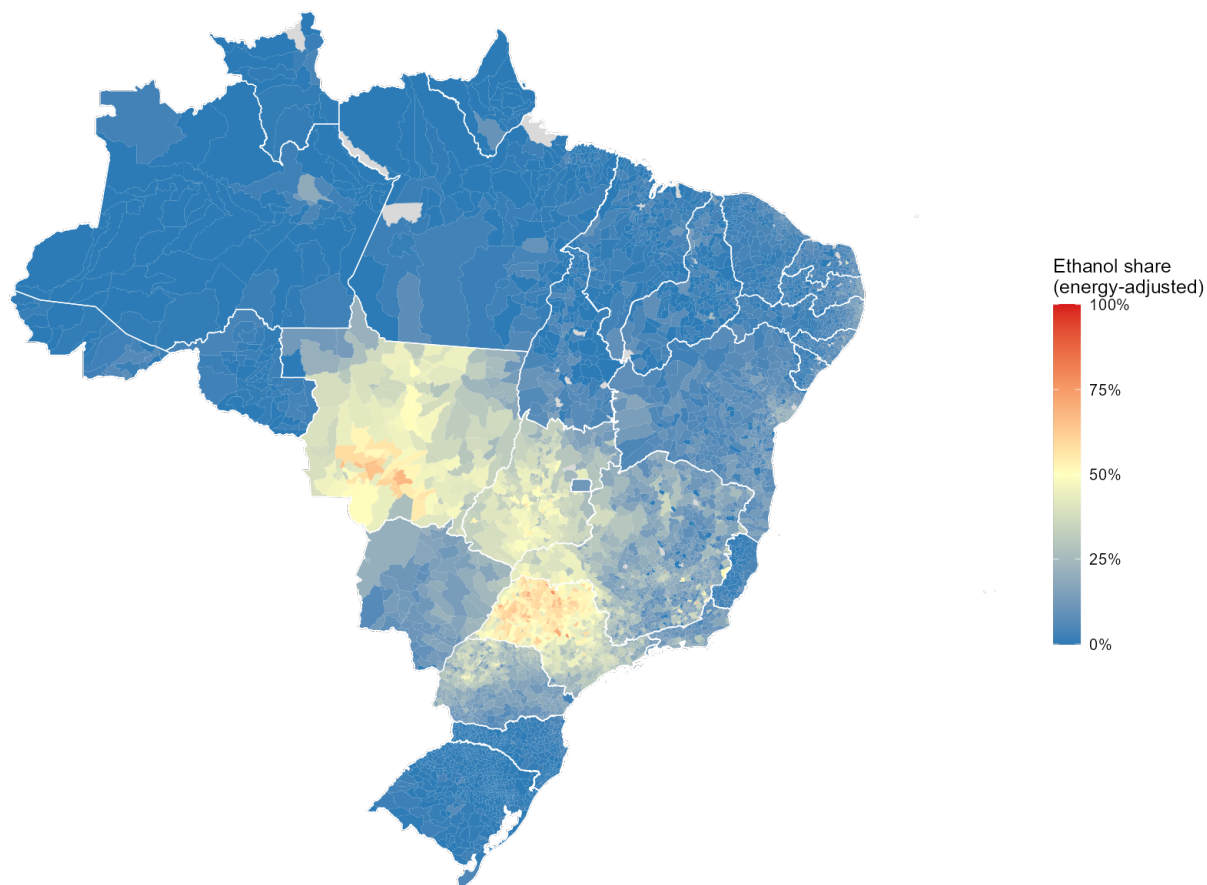
Figure A-8: Sugarcane burned area in São Paulo by season, 2000–2023. Harvest season = July–September; off-season = January–March. Source: MODIS MCD64A1 monthly burned-area product crossed with MapBiomas LULC sugarcane layer (Collection 9), aggregated across all São Paulo municipalities via Google Earth Engine.

Figure A-9: Distribution of Sugarcane Harvesting Methods in São Paulo State (Mechanical vs. Manual)



Data from Conab ([Conab, 2024](#)). The figure illustrates the distribution of harvested sugarcane in São Paulo by harvesting method.

Figure A-10: Energy-Adjusted Hydrated Ethanol Share in the Fuel Mix in Brazil (2020–2024)



Notes: Share defined as $V_{eth} \times (1/1.4) / [V_{eth} \times (1/1.4) + V_{gas}]$, where $1/1.4 \approx 0.714$ is the energy equivalence factor (EPE/ANP standard): one liter of gasoline C contains the same energy as 1.4 litres of hydrated ethanol. Municipal averages computed over 2020–2024. Municipalities with fewer than three years of data are excluded (shown in grey). Source: ANP — *Vendas Anuais de Derivados de Petróleo e Biocombustíveis por Município*.

Figure A-11: Event Study — Respiratory Hospitalizations (Full Sample)

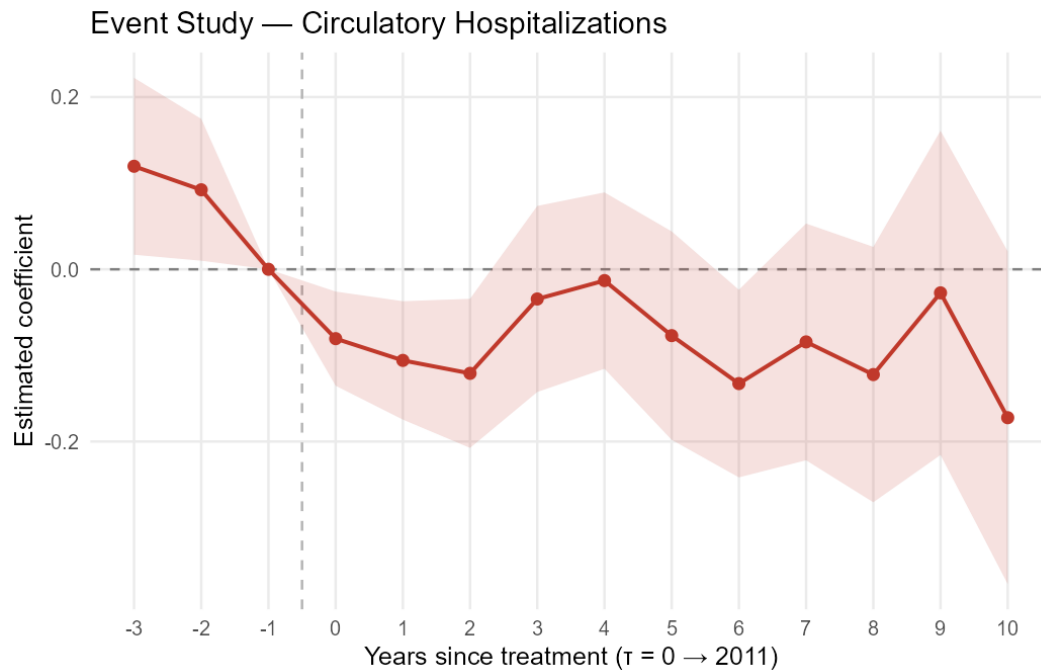
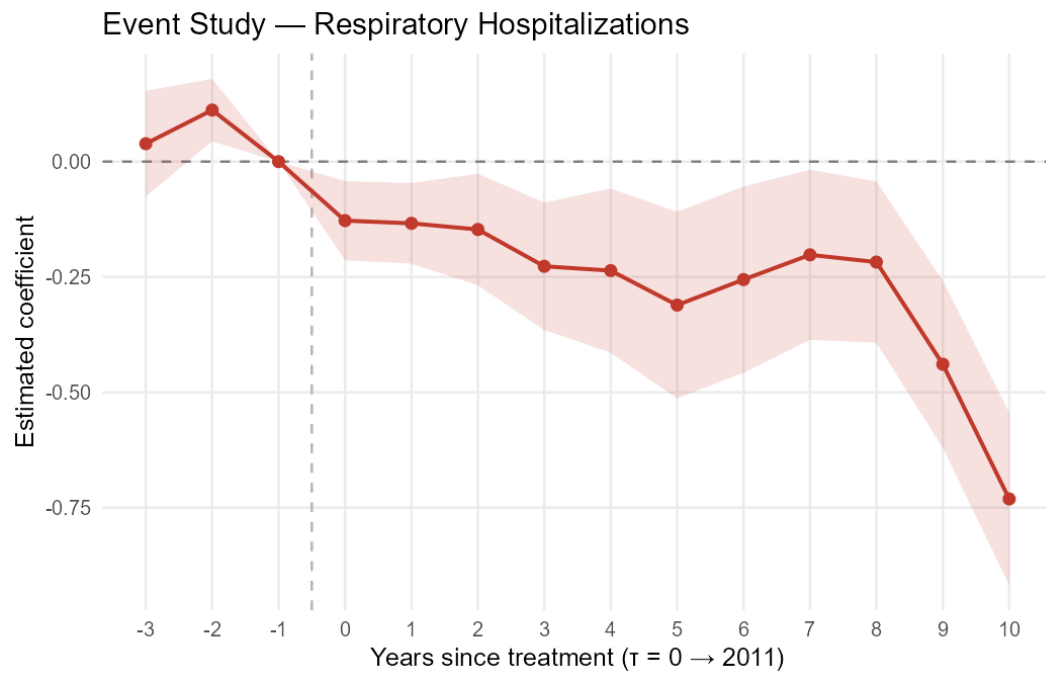
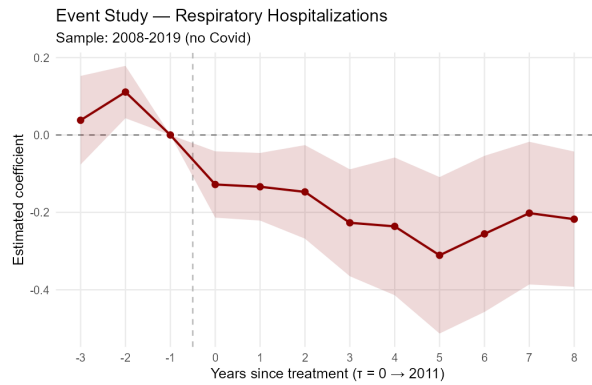
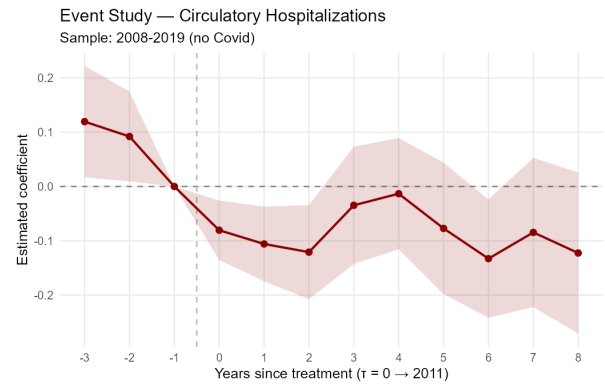


Figure A-12: Event Study — Circulatory Hospitalizations (Full Sample)



(a) Respiratory (Pre-Covid)



(b) Circulatory (Pre-Covid)

Figure A-13: Event Study — Pre-Covid Sample (2008–2019)

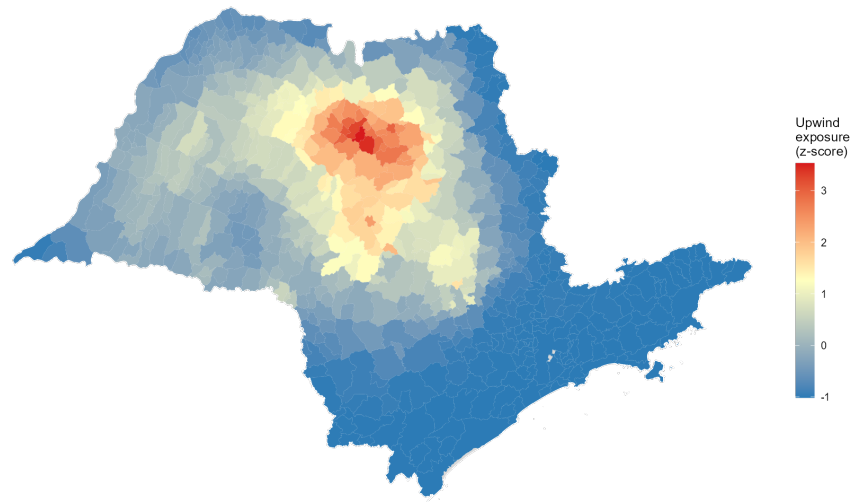
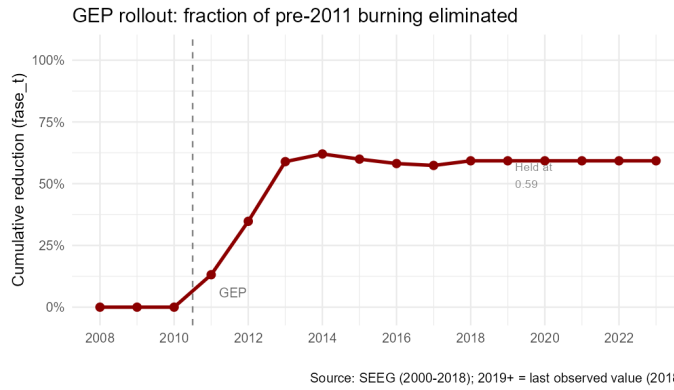


Figure A-14: Upwind Sugarcane Exposure by Municipality (*WindExp*, harvest season average)

Figure A-15: Continuous Treatment Intensity (f_{ase_t}): Fraction of Pre-2011 Sugarcane Burning Emissions Eliminated



f_{ase_t} from SEEG (2000–2018); held at last observed value (0.59) for 2019 onward.

Table A-1: Birth Outcomes — Municipality-Year Aggregate

	$\ln(\text{birthweight})$	$\ln(\text{lbw})$	$\ln(\text{vlbw})$	$\ln(\text{anomalies})$
<i>Panel A — Full sample (2002–2023)</i>				
Exp	−0.003 (0.004)	0.029 (0.023)	−0.064 (0.058)	−0.844*** (0.319)
Observations	15,484	15,485	15,485	9,766
<i>Panel B — Pre-Covid (2002–2019)</i>				
Exp	−0.003 (0.004)	0.029 (0.023)	−0.025 (0.055)	−0.836** (0.326)
Observations	12,905	12,906	12,906	8,163
Std. Errors by municipality	✓	✓	✓	✓
Municipality FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Weights	Births	Births	Births	Births

** $p < 0.05$, *** $p < 0.01$. Municipality-year aggregate outcomes. Outcomes: $\ln(\text{mean birthweight})$, $\ln(\text{share LBW} + 0.001)$, $\ln(\text{share VLBW} + 0.001)$, $\ln(\text{number of anomalies})$. Compare with Table 2 for individual-level estimates.

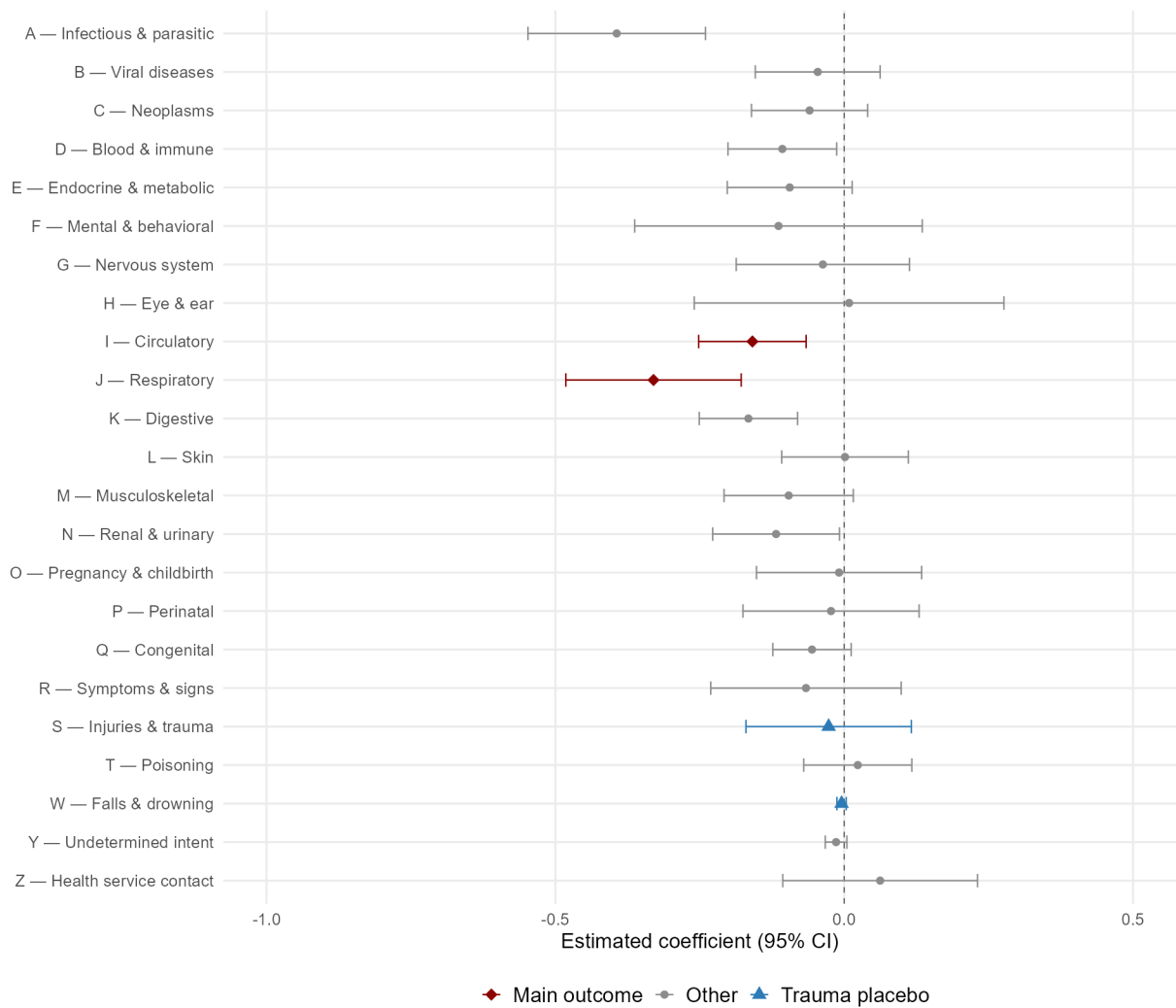


Figure A-16: Effect of sugarcane mechanization across all ICD-10 chapters.

Notes: Each point is $\hat{\beta}^c$ from estimating $\ln(\text{hosp. rate}_{it}^c + 0.001) = \alpha_i + \alpha_t + \beta^c \cdot TC_{it} + \varepsilon_{it}$, where $TC_{it} = Land_i \times D_{t>2010}$. Municipality and year fixed effects throughout; standard errors clustered at the municipality level; weighted by population. Data: SIH/DataSUS via Base dos Dados, São Paulo, 2008–2021; hospitalizations assigned to the patient’s municipality of residence.

Table A-2: Robustness: Alternative Pre-Treatment Averaging Windows

	Pre-treatment averaging window for sugarcane land share		
	2000–2010	2005–2010	2008–2010
<i>Panel A — Respiratory Diseases (ICD J00–J99)</i>			
Tech. Change (α)	−0.386*** (0.093)	−0.368*** (0.087)	−0.343*** (0.083)
SD(exposure)	0.179	0.193	0.208
Effect at 1 SD	−6.9%	−7.1%	−7.1%
Observations	8,988	8,988	8,988
<i>Panel B — Circulatory Diseases (ICD I00–I99)</i>			
Tech. Change (α)	−0.178*** (0.055)	−0.173*** (0.052)	−0.163*** (0.050)
SD(exposure)	0.179	0.193	0.208
Effect at 1 SD	−3.2%	−3.3%	−3.4%
Observations	8,988	8,988	8,988
Municipality FE	✓	✓	✓
Year FE	✓	✓	✓
Weights	Pop.	Pop.	Pop.

Standard errors clustered by municipality in parentheses. *** $p < 0.01$. The dependent variable is the logarithm of hospitalizations per capita, estimated from equation (1). Each column recomputes the treatment variable – the municipal sugarcane land share – as the average over the indicated pre-2011 window, holding the rest of the specification fixed. “Effect at 1 SD” reports $\hat{\alpha} \times SD(\text{exp})$, the implied percentage change in hospitalizations from a one-standard-deviation increase in exposure under each window. Sample: São Paulo municipalities, 2008–2021.

Table A-3: Robustness: Minimum Sugarcane Share Required for Exposure

Minimum share c	Respiratory		Circulatory		Exposed municipalities
None (baseline)	-0.386***	(0.093)	-0.178***	(0.055)	553
1%	-0.386***	(0.093)	-0.178***	(0.055)	437
3%	-0.387***	(0.093)	-0.179***	(0.055)	389
5%	-0.385***	(0.093)	-0.176***	(0.055)	349
6.7%	-0.377***	(0.091)	-0.172***	(0.055)	320
10%	-0.368***	(0.090)	-0.168***	(0.055)	275
15%	-0.357***	(0.089)	-0.151***	(0.055)	226

Standard errors clustered by municipality in parentheses. *** $p < 0.01$. Each row re-estimates column (1) of Table 1 with $TC_{it} = Land_i \times D_{t>2010} \times \mathbf{1}(Land_i > c)$, so that municipalities at or below the minimum share c are treated as unexposed. Hospitalizations by patient's municipality of residence, 2008–2021, weighted by population.

Table A-4: Robustness: Dose-Response by Quartile of Sugarcane Land Share

Post-2010 $\times$ quartile of sugarcane share	Q1	Q2	Q3	Q4
Mean sugarcane share in quartile	0.005	0.055	0.175	0.465
Respiratory hospitalizations	0.002 (0.054)	-0.034 (0.057)	-0.138*** (0.049)	-0.171*** (0.045)
Circulatory hospitalizations	0.025 (0.031)	-0.019 (0.036)	-0.080*** (0.022)	-0.063* (0.034)

Standard errors clustered by municipality in parentheses. * $p < 0.1$, *** $p < 0.01$. Each row is a single regression of the outcome in Table 1, column (1), on post-2010 indicators for each quartile of sugarcane land share, with municipality and year fixed effects, weighted by population. Quartiles split the 553 municipalities with positive share (cutoffs at shares of 0.019, 0.098, and 0.271); the 89 municipalities without sugarcane are the reference group. Hospitalizations by patient's municipality of residence, 2008–2021. Terciles and quintiles show the same pattern: no effect in the lowest groups and reductions of about 0.13–0.17 in respiratory hospitalizations in the upper groups.

H.1 Robustness: Excluding the 2009 H1N1 Influenza Pandemic Year

The pre-treatment test of parallel trends (Section 6.1, Figure A-11) reveals a spike in respiratory hospitalizations in 2009 (event-study coefficient = +0.112, SE = 0.035), coinciding with the 2009 H1N1 influenza pandemic (“swine flu”) in Brazil – an episode unrelated to, and a decade earlier than, the COVID-19 pandemic. This external health shock may confound our identification strategy in that year. To assess robustness, we re-estimate our main specifications excluding 2009 (Table A-H1).

Results show little sensitivity to excluding the H1N1 influenza year: the coefficient on *TC* in respiratory hospitalizations changes from -0.3567 (SE = 0.1062) with 2009 included to -0.3258 (SE = 0.1080) when 2009 is excluded—a difference of only 0.0309. This stability is reassuring given that our policy-relevant window is post-2011 (after technical change implementation); the pre-2011 period serves as background control variation, and a single-year external shock does not materially alter treatment-effect estimates.

For SINASC birth outcomes, we note elevated congenital anomalies in 2009 (8.46 per 1,000 births) compared to 2008 (7.35), consistent with documented H1N1 maternal infection effects ($p = 0.045$). The *TC* coefficient in the full model is -0.4638 (SE = 0.2378); excluding 2009 yields -0.4099 (SE = 0.2671). The estimate keeps its sign and magnitude, but it is only marginally significant with 2009 included and loses significance when 2009 is excluded, limiting strong inference. We interpret birth-outcome results with appropriate caution.

Table A-5: Robustness: Excluding 2009 (H1N1 Influenza Pandemic Year)

	With 2009		Excluding 2009	
	Coef	SE	Coef	SE
Respiratory hospitalizations (N observations)	-0.3567 8988	0.1062	-0.3258 8346	0.1080
Congenital anomalies (N observations)	-0.4638 9030	0.2378	-0.4099 8385	0.2671

Standard errors clustered by municipality in parentheses. Coefficients are on *TC* (Tech. Change indicator). Respiratory outcome: $\ln(\text{hosp. count} + 0.001)$ by patient’s municipality of residence, municipality $\times$ year level, 2008–2021. Birth outcome: $\ln(1000 \times \text{anomalies/births} + 0.001)$, municipality $\times$ year level, 2008–2021. Both regressions are weighted by municipal population.

Table A-6: Summary Statistics

Variable	<i>N</i>	Mean	SD	Min	Max
<i>Panel A — Treatment variable (municipality level)</i>					
Sugarcane land share (2000–2010)	642	0.150	0.190	0.000	0.886
<i>Panel B — Hospitalization outcomes (municipality × year, 2008–2021)</i>					
Respiratory hosp. per 1,000 pop. (I00–I99)	8,988	7.784	5.198	0.277	46.398
Circulatory hosp. per 1,000 pop. (I00–I99)	8,988	8.107	3.980	0.603	38.560
<i>Panel C — Birth outcomes (municipality × year, 2002–2023)</i>					
Any congenital anomaly (per 1,000 births)	14,189	18.662	91.807	0.000	1,000
Pol3 index: NTD, orofacial & cardiac (per 1,000 births) ^a	14,189	3.590	59.809	0.000	1,000
Low birthweight <2,500 g (%)	14,189	9.341	9.510	0.000	100.000
Very low birthweight <1,500 g (%)	14,189	1.550	5.068	0.000	100.000
<i>Panel D — Mortality outcomes (municipality × year, 2001–2023)</i>					
Infant resp. mortality, <5 years (per 1,000 births)	14,765	0.832	3.343	0.000	111.111
Resp. mortality, all ages (per 100,000 pop.)	14,766	89.507	47.575	0.000	546.448
Circ. mortality, all ages (per 100,000 pop.)	14,766	201.490	73.015	0.000	718.993

Notes: Unit of observation is the municipality × year for Panels B–D and the municipality for Panel A. Hospitalization rates from SIH/DataSUS (2008–2021), by patient’s municipality of residence; birth outcomes from SINASC (2002–2023); mortality from SIM (2001–2023), by municipality of residence. São Paulo state municipalities only. Large max values in Panels C and D reflect municipality-year cells with very few births or deaths; all DiD regressions use population or birth weights.

^a Pol3 is an individual-level binary indicator for births presenting at least one of: neural tube defect (Q00–Q07), orofacial cleft (Q35–Q38), or congenital cardiac malformation (Q20–Q26). The *N* shown is total live births; mean and SD reflect the individual-level binary distribution. All other rows in Panel C are municipality × year cell averages.